



THE UNIVERSITY OF
SYDNEY

—
**Sydney
Informatics Hub**

*BIO101 workshop:
Design Foundations for Omics Studies*

17th September 2026

We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.



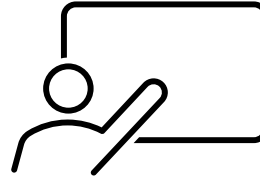
Celebrating 175 *years*

Welcome



Housekeeping

- Event runs **9am-4pm**
- 4 sessions, 3 breaks
- Coffee and tea provided
- Bathrooms are past the lifts



Delivery

- 3 learning modules
- Short talks, hand-on activities, and discussions
- Working in groups
- No coding today



Materials

- <https://sih.tools/bio101-workshop>
- Content will be emailed to you after the workshop

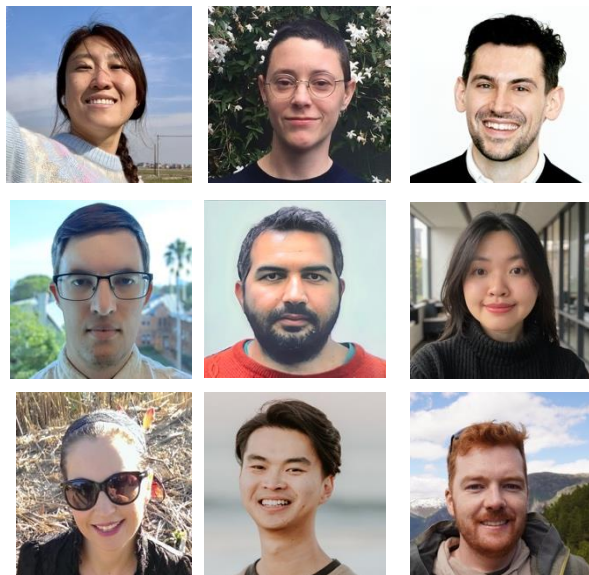
Meet the team

Trainers: Amarinder and Fred

Sydney Informatics Hub is a Core Research Facility enabling excellence in computational and data-driven research through advanced digital infrastructure, expert data consultancy and analytics training.



Bioinformatics @ SIH



We provide advice, data processing and analytical support, grant support, research computing platform onboarding, and training.



We support fully-subsidised national services that make it easier for you to apply bioinformatics to your research.

Find our resources at:

- sih.tools/github-external
- workflowhub.eu/projects/43
- sih.tools/bio-training
- biocommons.org.au/events
- biocommons.org.au/bioshell

Workshop housekeeping

Today's workshop is a combination of:

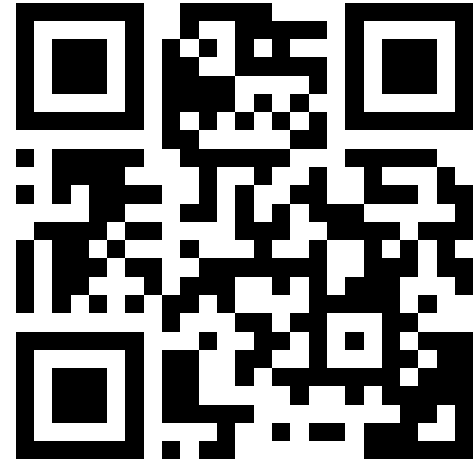
- Short lectures introducing concepts
- Discussions and quizzes
- Hands-on activities

All our materials are online:

- <https://sih.tools/bio101-workshop>

Need help?

- No such thing as a silly question
- Trainers will check in throughout the workshop
- Feel free to ask a question



Agenda

09:15-09:30	Workshop introduction
09:30-10:30	Module 1: Omics layers and the experimental workflow
10:30-10:45	A short break
10:45-12:00	Module 2: Designing robust experiments
12:00-12:45	Lunch
12:45-14:00	Module 2: Designing robust experiments
14:00-14:15	A short break
14:15-15:15	Module 3: Apply your learning
15:15-16:00	Wrap up and Q&A

Activity: introduce yourself

Which organism do you primarily work with?

- 1. Human**
- 2. Plants**
- 3. Animals**
- 4. Microbes**
- 5. Communities**

What is your primary research or application area?

- 1. Health and medicine**
- 2. Agriculture and food production**
- 3. Wildlife and conservation**
- 4. Environmental modelling**
- 5. Fundamental research**

How much experience do you have working with omics data?

- 1. Never used omics data**
- 2. Interpreted or used omics results but didn't generate them**
- 3. Involved in study design or data generation**
- 4. Regularly design and/or analyse omics experiments**

Which molecular layer are you working with or most curious about?

- 1. DNA (genomics, epigenomics)**
- 2. RNA (transcriptomics)**
- 3. Protein (proteomics)**
- 4. Metabolites (metabolomics)**
- 5. Multiple or not sure**

Module 1.1: The omics landscape

Module 1.1 learning objectives

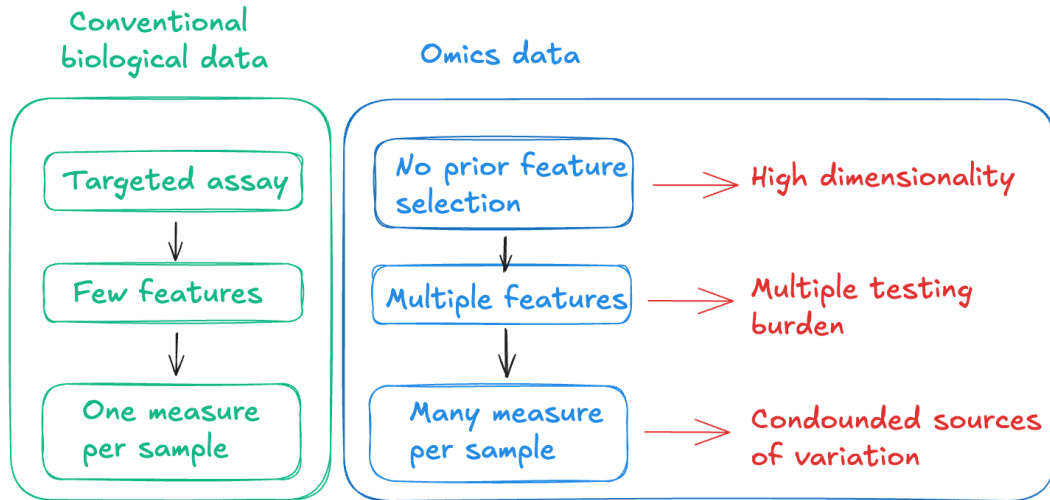
- Describe the molecular information captured by each of the 5 omics layers
- Explain how the molecular layers relate through the central dogma and its regulatory steps
- Select the most appropriate molecular layer for a given biological question
- Evaluate the limitations of each layer and what a chosen layer cannot tell you

What does “omics” mean?

The characterisation and quantification of sets of biological molecules and the investigation of how they translate into structure, function, and dynamics of an organism or group of organisms.



What are the unique features of omics data?



High dimensionality: measure far more features than you've sampled which violates lots of classical statistical method assumptions.

Multiple testing burden: each additional feature tested is another opportunity for a false positive.

Confoundered sources of variation: Technical and biological variation are both found in omics data.

The layers: from DNA to metabolite

What are the molecular layers?

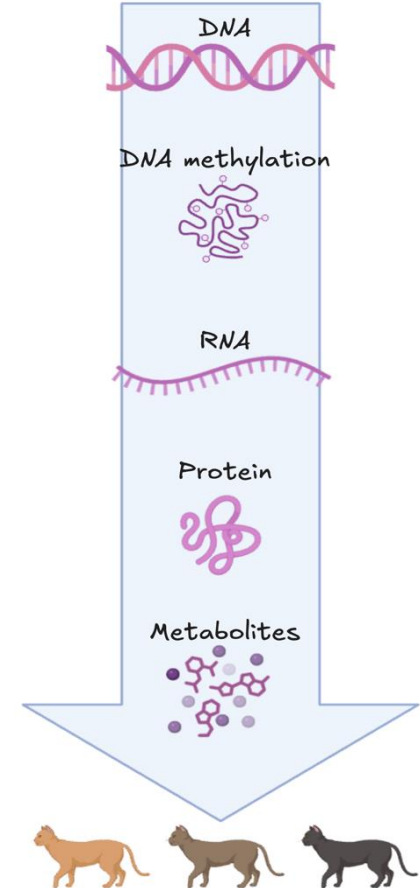
- Each is a distinct class of biological molecules
- Ordered by their position in the Central Dogma
- They are connected but not redundant

What information do they capture?

- Each layer captures a different dimension of biological state

Why do they matter in study design?

- Determines the biological processes it can detect
- Determines your analytical framework
- Determines what research questions you can answer



Layer 1: DNA (the genome)

What is it?

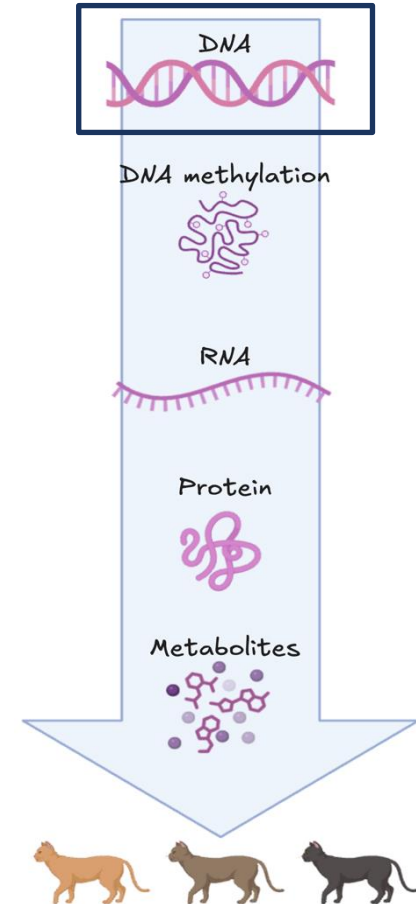
- The complete set of DNA in an organism. It encodes genes and regulatory elements

What is its role in biology?

- A stable repository of hereditary information
- Determines which genes an organism possesses, how they are structured, and regulatory logic that governs downstream molecular layers

What can it tell me?

- Genes and genetic variation present and their structure
- Comparative questions for populations, evolutionary processes



Layer 2: DNA modification (the epigenome)

What is it?

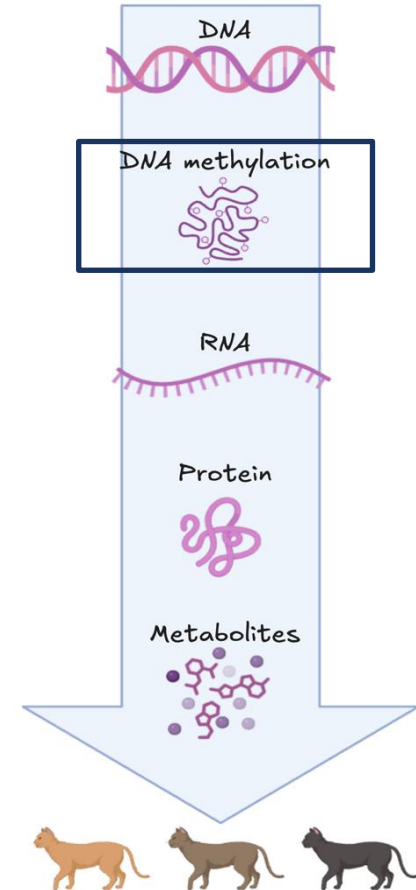
- Chemical modifications to DNA and histone proteins that determine which regions of the DNA is accessible

What is its role in biology?

- Regulatory interface between the genome and its dynamic environment
- Allows different cells to express different genes

What can it tell me?

- Regulatory state of the genome in a given cell type
- Functional relevance of variants in non-coding parts of the genome



Layer 3: RNA (the transcriptome)

What is it?

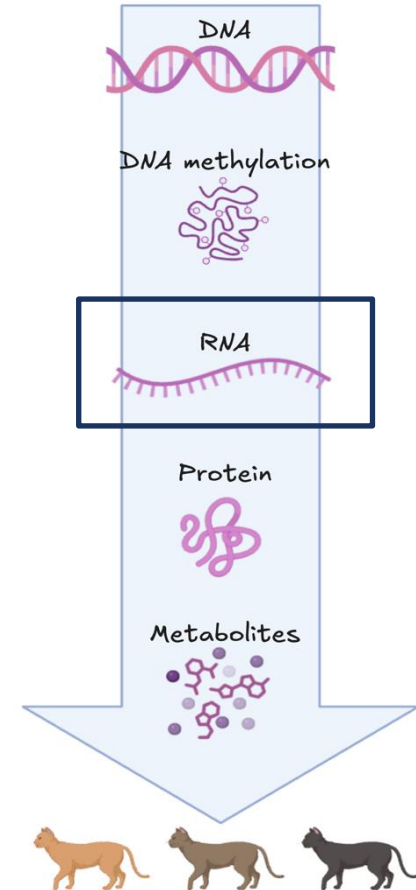
- Complete set of RNA molecules produced in a cell or tissue at a given time

What is its role in biology?

- Gene expression is the primary mechanism by which cells regulate their functional state in response to changing conditions

What can it tell me?

- Which genes are active, at what level, and in which isoform



Layer 4: Protein (the proteome)

What is it?

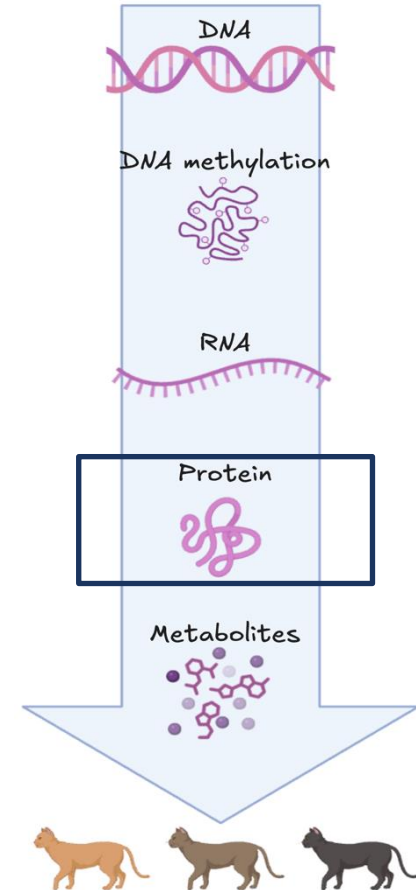
- The complete set of proteins present in a cell or tissue at a given time
- Proteins are the primary functional molecule of the cell and often function as multi-protein complexes

What is its role in biology?

- Execute virtually all cellular functions
- Set of molecular machines that carry out instructions encoded in upstream processes

What can it tell me?

- Which proteins are present and at what abundance (but not if they're active)



Layer 5: Metabolite (the metabolome)

What is it?

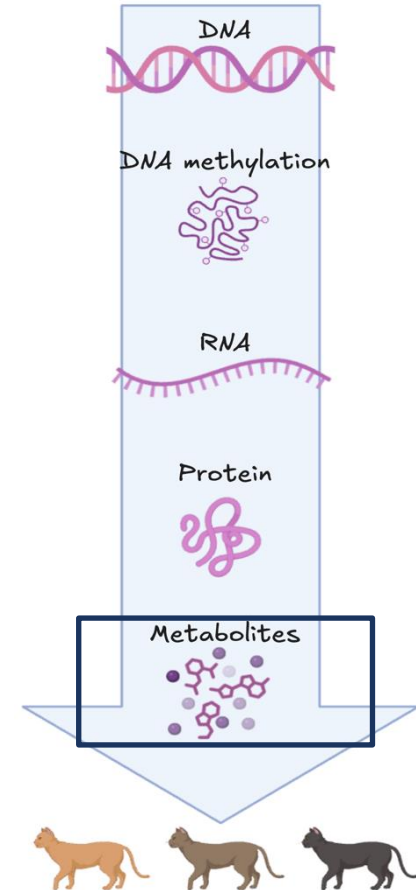
- The complete set of small molecules produced, consumed, and modified during cellular metabolism
- Includes sugars, lipids, nucleotides, and organic acids

What is its role in biology?

- Substrates and products of every enzymatic reaction in the cell
- Integrates information from all upstream layers and the environment

What can it tell me?

- Which metabolic pathways are active, the cell's energetic status, and where metabolic blocks have arrived



Most questions can be approached from more than one layer

Takeaway: pick the molecular layer(s) where your signal is the dominant source of variation, not one where it competes with sources of variation you are not studying.

Discussion: applying the molecular layers



Discussion: what is going on here?

A population of Maine Coon cats are screened at a vet clinic. Some have severe hypertrophic cardiomyopathy (HCM). Some have a **known MYBPC3 variant** but no cardiac abnormality. Some have HCM with no identified variant. Same breed, same environment, different outcomes.

What is driving cardiac disease in this population?

What can each molecular layer tell you about what is driving their heart failure, and what would it leave unanswered?

Activity: applying the molecular layers



Group activity: apply the layers

1. Form groups based on your answers to the welcome question about organism you work with
 - human 😊, microbe 🔬, animal 🐶, plant 🌱
1. Choose one of the examples on the table to work on together
2. Answer the questions as a group on your butchers paper:
 - **Which molecular layer(s) is suited to answering your question?**
 - **What can this layer tell you, what might it miss?**
3. Report back to the group



Group activity: production biology scenario

Why do some farmed fish grow faster than others despite receiving the same diet?

In a commercial Atlantic salmon cohort, fish are reared under identical conditions yet body mass diverges substantially within a single cohort over a grow-out cycle. By harvest, the heaviest individuals may be twice the mass of the lightest, with no obvious behavioural or health explanation.

Skeletal muscle growth in fish is regulated by the GH/IGF-1 axis, with myostatin acting as a negative regulator of muscle mass (the same core pathway as in mammals). Nutrient sensing through mTOR integrates amino acid availability with anabolic signalling. Feed conversion efficiency (how much dietary protein is deposited as muscle versus oxidised) varies between individuals and is partly heritable in salmonids. Early thermal or nutritional experience during embryogenesis can programme lasting differences in metabolic phenotype through mechanisms that persist without changes to the underlying DNA sequence.

Liver, skeletal muscle, gut tissue, and blood are all accessible. Individual fish can be tagged and tracked longitudinally, allowing repeated blood sampling from the same animal, though tissue biopsies are terminal.



Group activity: infectious disease scenario

Why do some populations tolerate an infectious disease while others suffer severe disease from the same pathogen?

Chytridiomycosis, caused by a fungal pathogen has driven amphibian population collapses globally. Yet exposure to the same pathogen produces markedly different outcomes across and within species: some populations suffer near-total mortality, others show subclinical infection and persist, and a small number of previously decimated populations appear to be recovering. This suggests that tolerance or resistance can emerge over ecological timescales.

This fungal pathogen targets keratinocytes in the skin. In susceptible hosts, infection disrupts cutaneous electrolyte transport causing osmotic imbalance that ultimately leads to cardiac arrest. MHC class II diversity in hosts has been associated with resistance in some species, and population bottlenecks following epidemic sweeps may have altered allele frequencies at immune loci. On the pathogen side, some isolates vary in virulence and strains differ in protease secretion and in the degree to which they suppress host immune signalling.

Skin swabs, skin punch biopsies, blood, and tissue from the granular gland layer are all collectable. Some populations can be sampled longitudinally; others are accessible only at single time points due to rarity or access constraints.



Group activity: clinical scenario

What is driving these patients' heart failure?

A cohort of patients presents to hospital with heart failure of unknown aetiology. Echocardiography shows reduced ejection fraction and dilated ventricles, but standard clinical workup with imaging, blood panels, and coronary angiography, has not identified a cause. Left ventricular biopsies are collected at the time of surgical intervention and stored. A subset of patients have first-degree relatives with similar presentations, suggesting a possible heritable component; others have no family history.

At the cellular level, cardiomyocytes in failing hearts are known to undergo transcriptional reprogramming (reactivating a "fetal gene programme") and to shift their metabolic substrate preference from fatty acids toward glucose. Sarcomeric proteins may carry inherited variants that alter contractile force. Mitochondrial dysfunction is common but whether it is cause or consequence is often unclear. The same clinical endpoint (pump failure) can arise from fundamentally different molecular origins.

Biopsy-derived material is limiting; matched blood samples and, for some patients, germline DNA from saliva are also available.



Group activity: environmental stress scenario

How does a crop plant respond to acute environmental stress, and what makes some varieties more resilient than others?

Three wheat varieties are exposed to a controlled drought stress regime in glasshouse conditions. Physiological measurements confirm that stress was imposed and that varieties differ in the severity of their response. One variety recovers quickly after rewatering; the other two show sustained suppression of growth.

Plant stress responses operate across multiple timescales. Within minutes to hours, stomata close in response to abscisic acid (ABA) signalling, and reactive oxygen species (ROS) accumulate as a stress signal. Over hours to days, transcriptional reprogramming occurs: stress-responsive transcription factors activate downstream genes involved in osmolyte synthesis, heat shock proteins, and antioxidant enzymes. At the metabolic level, compatible solutes such as proline and glycine betaine accumulate to maintain osmotic balance, and carbon metabolism shifts as photosynthesis is constrained. Underlying all of this is the genetic architecture of the varieties themselves with heritable differences in regulatory sequences, gene copy number, or structural variants that affect stress pathway components.

Leaf tissue is the primary sampling target but is metabolically dynamic as composition changes rapidly after excision, so processing time matters. Root tissue captures a different aspect of the drought response. Both stressed and recovering time points are available, as well as an unstressed baseline. The varieties differ in breeding history and are not isogenic lines, meaning genetic background differences exist alongside any stress-responsive differences.

Module 1.2: The omics study workflow

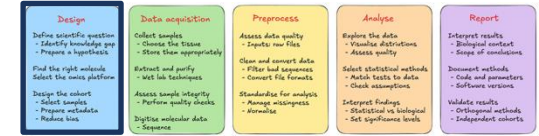
Module 1.2 learning objectives

- Identify the five stages of an omics study and the key decisions made at each stage
- Explain how methodological choices at one stage constrain what is possible downstream
- Recognise common sources of bias, technical variability, and analytical error in omics workflows
- Apply key design, acquisition, and analysis considerations to their own research context

Module 1.2 Omics Study Design workflow



Stage 1: design foundations



- 1. Scientific question: scope and purpose of the study**
- 2. Hypothesis: what is expected and at which molecular layer**
- 3. Variables: what is being compared and what needs to be considered**
- 4. Cohort structure: which biological units and groups will be included**
- 5. Platform: what the experiment can actually measure**

Stage 1: scientific question



You must turn a biological question into a study that can provide meaningful evidence.

Criterion	Factors to consider
What biological material and population are being studied?	Species; tissue and/or cell type; phenotypic status; age; sex; ancestry; sample availability; collection context
Which conditions, exposures, or comparison are being investigated?	Primary variables (disease, treatment, trait); comparison structure (case-control, longitudinal, single-condition); known or likely confounders
Which molecular layer could provide a relevant answer?	Known biology of the phenotype or process; what has been studied; required resolution; platform availability

Stage 1: hypothesis



Criterion	What it means for your hypothesis
Is the expected difference biological?	The hypothesis should predict a real biological state change, not an artefact of collection, processing, or measurement.
Is the direction or nature of difference specified?	A prediction of “difference” is insufficient. Your hypothesis should state what kind of change is expected.
Is it scoped to a specific molecular layer?	It should specify which molecular layer is expected to show change so that a null result at that layer is interpretable rather than redirected to another layer.
Is it bounded in biological context?	The hypothesis should be specific enough that a result in one tissue, population, or condition has clear scope and does not automatically generalise beyond it.

Stage 1: variables



Criterion	What it means for your experimental design
Have you identified your condition or exposure?	The primary biological variable driving the comparison must be defined before any other design decision can be made
Have you defined which omics features are relevant?	Knowing what you expect to change determines what needs to be measured e.g. genetic variants, gene expression, metabolite concentration, chromatin accessibility
Have you identified potential confounders?	Variables associated with the condition that could independently influence the omics features must be acknowledged
Have you decided how each confounder will be handled?	Depends on your circumstances. You can control it by design, balance it across groups, or at minimum record it in metadata

Stage 1: cohort structure



Criterion	What it means for your experimental design
Have you defined the independent biological unit?	The entity providing independent biological information, e.g. a participant, animal, or independently treated culture.
Do the samples represent the population you want to draw inferences about?	Samples specific to a context may not generalise. Findings are only interpretable relative to the population that your experimental cohort represents.
Do comparison groups differ only in the variable of interest?	Groups collected through different pathways carry systematic differences unrelated to the biology that may confound your results.
Have you identified and planned for known confounders?	Confounders should be controlled by design where possible and recorded in metadata so they can be evaluated during analysis
Does your cohort size provide adequate statistical power?	Sample size should ideally be determined by the expected effect size and platform variability, not by budget or availability alone

Stage 1: platform



Criterion	What it means for your experimental design
Does the platform measure the molecular layer you need?	The platform must be capable of capturing the molecule or feature the hypothesis predicts will change, not a proxy or an upstream correlate
Does the platform provide the required resolution?	Bulk, cell-type specific, or spatial, the resolution must match what the comparison requires
Can the platform detect the signal at the relevant abundance range?	Sensitivity and dynamic range determine whether the molecule of interest will be measured at all, not just whether differences between groups can be detected
Is the sample type compatible?	Fresh, frozen, or FFPE, sample availability constrains platform options before any other decision is made
Can the cost per sample support an adequately powered study?	Platform choice sets the upper bound on sample size under a fixed budget. If cost per sample is too high, the study cannot be powered regardless of design quality

Activity: experimental design audit

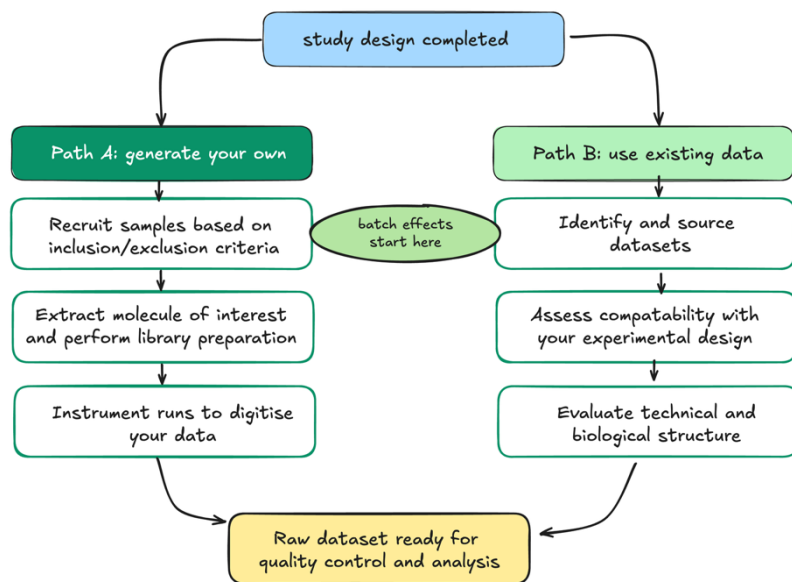


Improve your research question

Next to your research outline on the wall, add another sticky note, each explaining your intended approach using the following template:

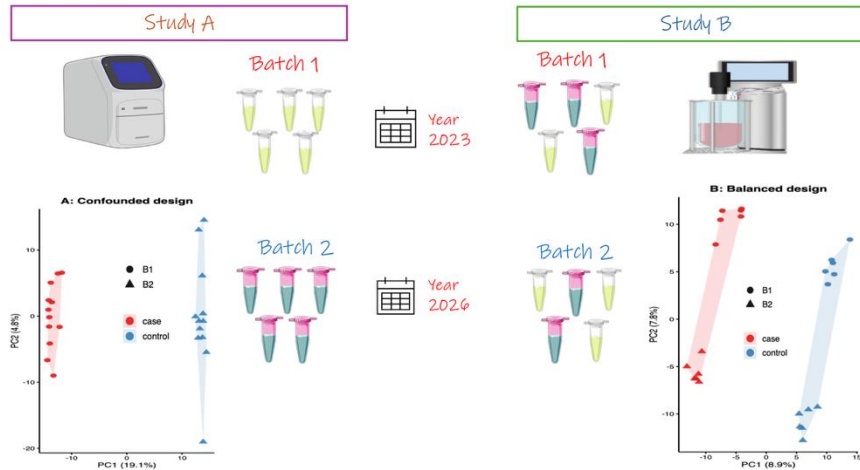
- 1. What am I measuring?**
- 2. In whom/what and why this population?**
- 3. What do I expect to find at my chosen molecular layer?**

Stage 2: data acquisition



	Path A	Path B
Benefits	Control over sample selection, collection protocol, processing conditions, and batch structure	Faster and lower cost; large existing cohorts may not be reproducible
Burdens	Time-intensive; recruitment, extraction, and sequencing happen in stages, introducing batch structure by default	Cohort, protocol, instrument, and pipeline are bundled into a single label you cannot separate
Risks	Batches that align with your biological comparison	Hidden confounding within the data
Advantages	You choose which samples go into which batch,	Excellent as an independent validation cohort,

Stage 2: batch effects and confounding



Batch effects are systematic technical bias introduced when samples are processed under different conditions.

Unlike random noise, they can produce consistent and reproducible patterns in the data that may be mistaken for biological variation.

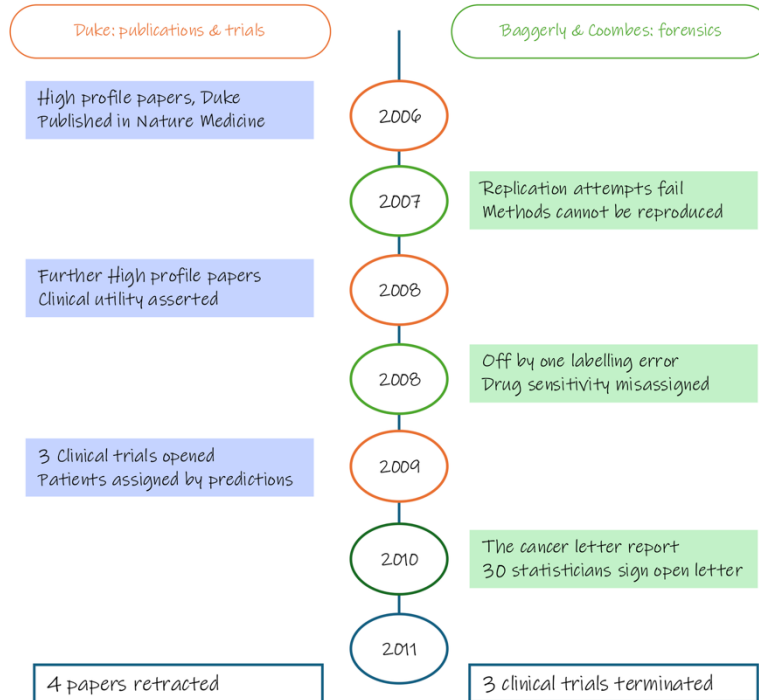
Stage 2: experimental controls



Experimental controls are known-composition samples processed alongside experimental samples to monitor technical variability, detect contamination, and support normalisation.

Control type	What is monitors	Example
Pooled QC or reference sample	Run-order drift; between-batch instrument variability	Aliquot of all study samples pooled, re-injected at regular intervals
Blank or negative control	Contamination; background signal	Reagent blank, extraction blank, solvent injection
Positive control	Assay sensitivity and specificity	Certified reference material; sample with verified signal
Internal standard or spike-in	Extraction efficiency; instrument response per sample	ERCC spike-ins (RNA-seq); iRT peptides (proteomics DIA); stable isotope-labelled metabolites

Stage 2: case study



The claim (2006-2008): Genomic signatures derived from cancer cell line drug-response data could predict which chemotherapy a patient would respond to.

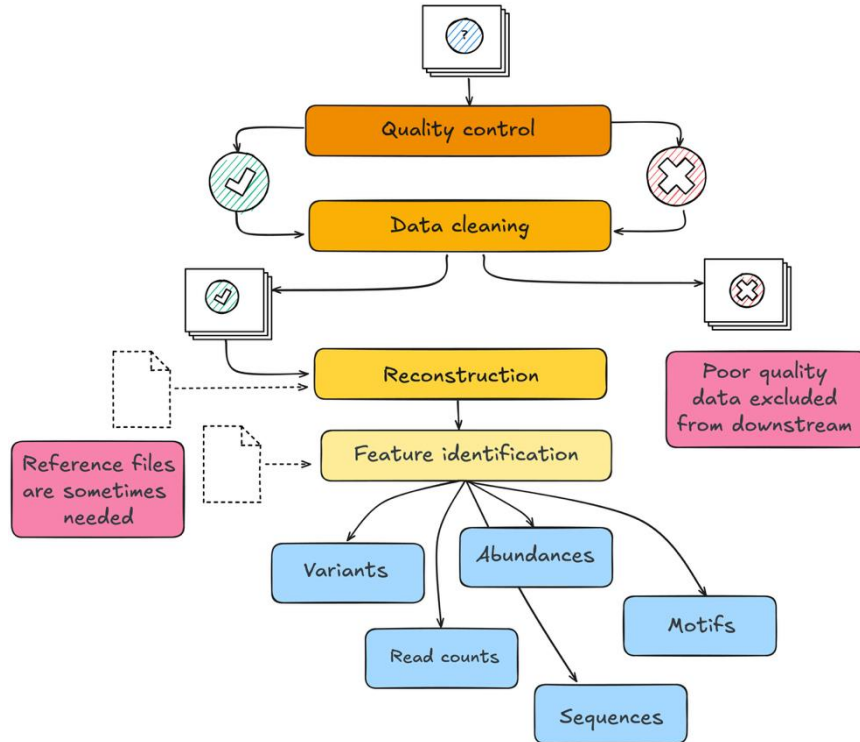
Forensic analysis (2007-2008):

- **Off-by-one indexing error** in sample labelling swapping which cell lines were classified as drug-sensitive vs drug-resistant
- **Batch confounding** of study origin with the biological label
- **Reproducible artefacts** mistaken for "predictive signatures"

The consequence (2009-2011):

- Three clinical trials opened based on the flawed predictions.

Stage 3: preprocessing



Raw instrument outputs are not directly suitable for biological analysis. They contain:

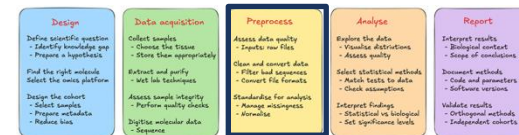
- Low-quality measurements
- Technical artefacts
- Contamination
- Systematic technical variation

Reconstruction and feature identification often require external reference files which determine what you can identify.

Features can be excluded at any stage. You may recover them by reprocessing from raw data!



Stage 3: quality control



Level	What can go wrong	Consequence
Sample	Failed extraction, degradation, contamination, low input	Noise and outliers in group comparisons
Run or instrument	Drift, instability, inconsistent acquisition	Systematic trend across all measurements in that run
Feature	Signal indistinguishable from background or blank	Inflated feature set with no biological contribution

Downstream analyses cannot distinguish bad data from biology, QC is your opportunity to catch it.

Are some samples outliers?

- Add noise or bias to the group's estimates

Is the signal quality sufficient?

- Misidentified features and noisy values

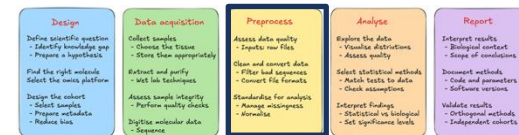
Is there contamination?

- Foreign signals attributed to samples

Are there technical artefacts?

- Instrument drift introduces trends indistinguishable from biological differences

Stage 3: data cleaning



Target	Common criteria	Risks if is wrong
Samples	Failed QC metrics, insufficient depth or coverage	Biased group composition if failures are systematic
Features	Low prevalence, low abundance, high missingness	Removes low-abundance signal that is biologically real
Values	Below detection limit, high technical variability	Structured missingness misread as random dropout

Filtering acts on what QC identifies. Use your filters to improve reliability without removing important biological signals.

Is the threshold based on your data or a default?

- Defaults are not always suited to your data

Does low prevalence mean biologically irrelevant?

- Low prevalence doesn't mean low importance

Is missingness random or structured

- Absent in one condition may be the biology, not a technical failure

Stage 3: reconstruction



Level	What can go wrong	Consequence
Reference	Outdated, incomplete, or mismatched reference	Reads or signals that don't match are not identified
Method	Wrong algorithm for the data type and shape	Wrong statistical model, features merged, split, or incorrectly resolved
Parameters	Thresholds and settings left at defaults	Results are sensitive to choices never made explicitly

Reconstruction translates cleaned signal into biological units. Methods and reference datasets determine what comes out.

Is the reference appropriate for your sample?

- Missing taxa, isoforms, gene models cannot be identified

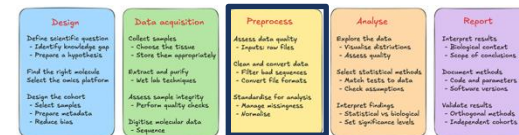
Does the method suit the data?

- Different statistical models have different assumptions and produce different outputs

Were parameters chosen explicitly?

- Default settings are usually quite reliable, but they weren't calibrated to your needs

Stage 3: feature identification



Level	What can go wrong	Consequence
Assignment	Signal matches multiple features equally	Discarded, arbitrarily assigned, or collapsed to a higher level
Quantification	Measured value doesn't reflect feature	Abundance estimate unreliable or misleading
Annotation	Feature identified but poorly characterised	Passes all thresholds; contributes nothing to interpretation

Feature identification assigns biological identity and quantifies the reconstructed signal. How ambiguous cases are handled determines which features are counted and at what abundance.

How are ambiguous assignments handled?

- Different rules produce different feature sets

Does the measured value reflect the feature?

- Detection limits, dynamic ranges, and saturation all determine whether metrics are reliable

Is annotation sufficient for interpretation?

- Functional characterisation is essential for interpretation

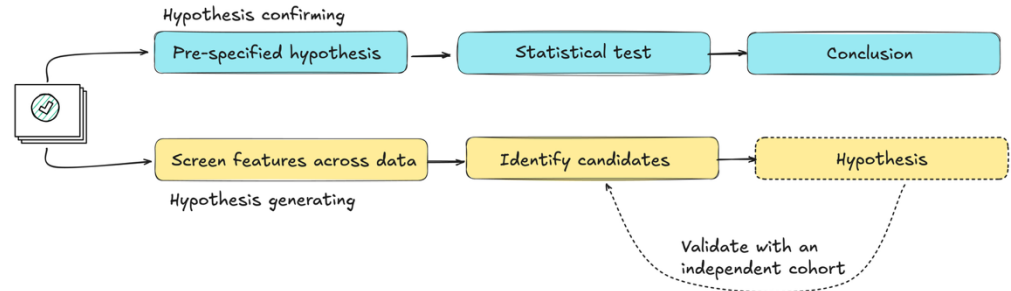
Stage 4: analysis



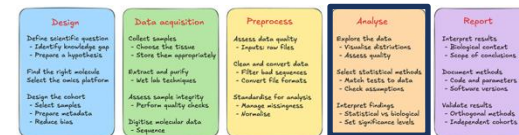
During analysis, we apply statistical models to pre-processed data to test our hypotheses and answer our research questions. The models we use must accurately reflect the structure of our data.

Omics studies can either confirm or generate a hypothesis:

- Confirming: does gene X differ between cases and controls?
- Generating: which of 20,000 genes differs between cases and controls?



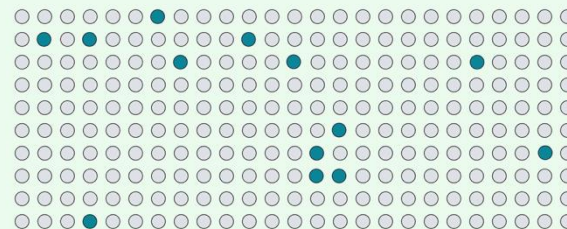
Stage 4: challenges of analysis at scale



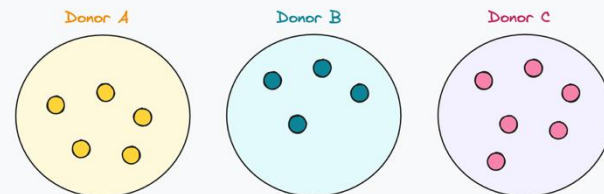
Two challenges require explicit design decisions:

1. Thousands of features are tested simultaneously
2. Observations are frequently not independent

At $p < 0.05$, one in twenty is expected significant by chance



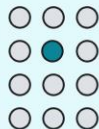
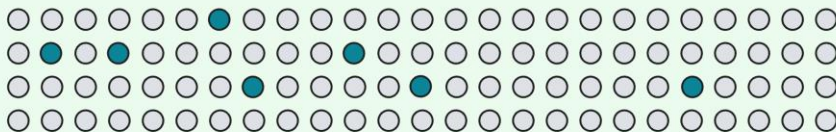
Measurements from the same biological unit share a common background



Stage 4: controlling for false positives



At $p < 0.05$, one in twenty is expected significant by chance

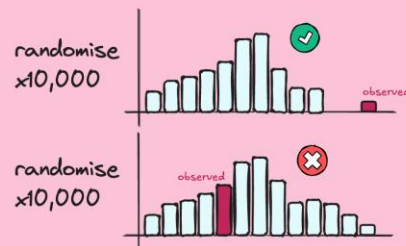


Solution 1: multiple testing correction

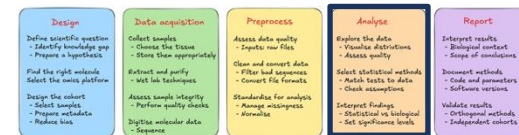
Adjust the significance threshold to account for the number of tests performed. This will control the expected proportion of discoveries that are false positives.

Solution 2: Permutation testing

Ask how extreme are the observed results relative to chance?



Stage 4: pseudoreplication case study



EXPERIMENTAL DESIGN

Cells from all 60 donors were pooled and analysed as if each cell were a statistically independent unit; ignoring that cells from the same patient share that patient's genetics, disease stage, age, and environment.

60 DONORS → ~80,000 CELLS



EFFECTIVE SAMPLE SIZE CLAIMED

$n \approx 80,000$
cells as independent observations

TRUE INDEPENDENT UNITS

$n \approx 60$ donors

CONSEQUENCE

AT FDR < 0.01

1,031

differentially expressed genes

vs correct analysis
(correct $n = 26$)

549x

inflation of gene discoveries

AT FDR < 0.05

14,274

differentially expressed genes

THE CONSEQUENCE GOES BEYOND THE NUMBERS

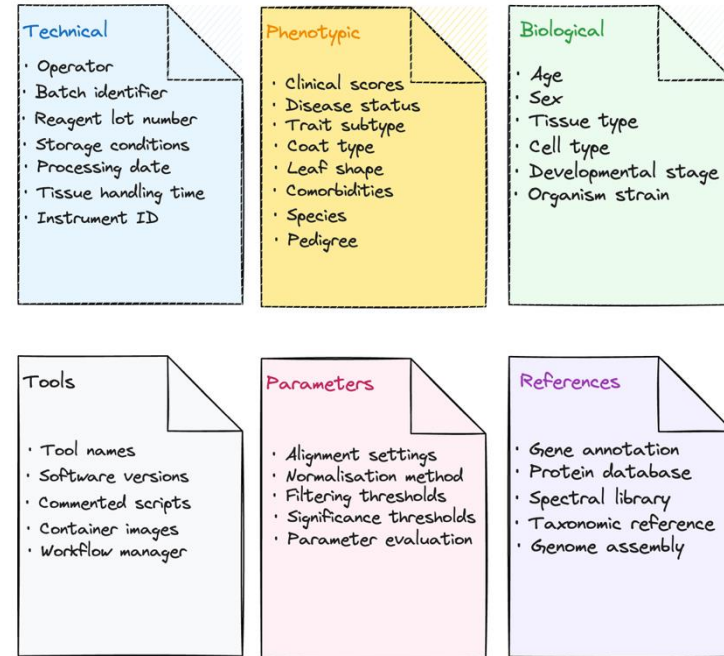
The inflated DEG list did not just over-report genes, it also pointed to the wrong cell type. The corrected re-analysis recovered a different, biologically correct signal.

Stage 5: reporting

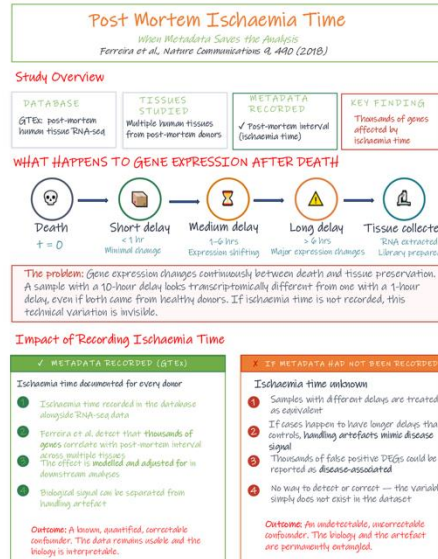


Two distinct types of information are required to reproduce a study:

1. **Metadata:** contextual information that makes a dataset interpretable
2. **Methods:** the computational record of every decision that shaped the analysis



Stage 5: metadata completeness



KEY MESSAGE

GTEx recorded ischaemia time, so the problem could be found and fixed.
 Most smaller studies do not.
 The difference between a recoverable and an unrecoverable confound is often simply whether someone thought to write it down at the time of collection.

✓ Record post-mortem interval for every donor

⚠ Missing metadata = permanent, invisible confound

Activity: apply the experimental workflow

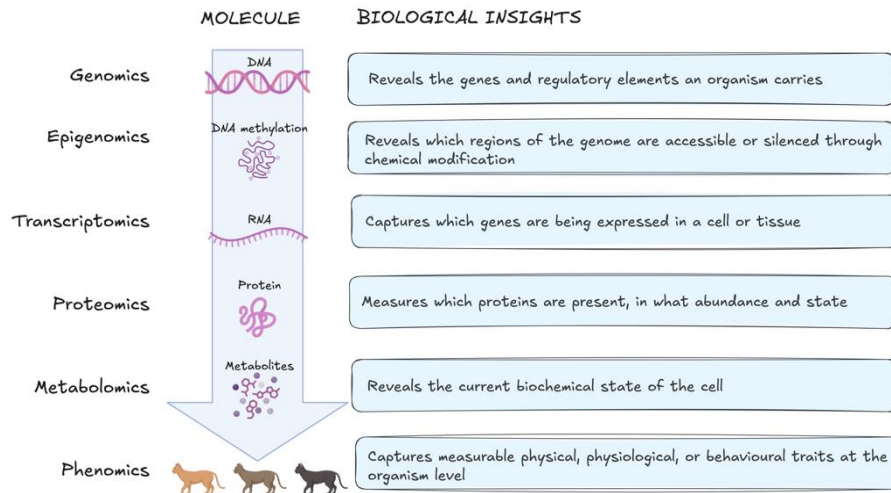


Map the scenario across the experimental workflow

Apply what you've learned about the 5 stages of an omics study to one of the scenarios on your table and answer the following questions:

1. Given the biological and logistical constraints described, which stage of the workflow will be hardest to plan, and what makes it difficult?
2. What would a reader need to know to reproduce the analysis and evaluate whether the findings are generalisable?

Module 1 takeaways



Experimental stage	Key question
1. Design	Are we comparing the right things?
2. Data acquisition	Can we measure them reliably?
3. Preprocessing	Can we identify reliable data?
4. Analysis	What signal is actually supported?
5. Reporting	What can we reasonably claim?

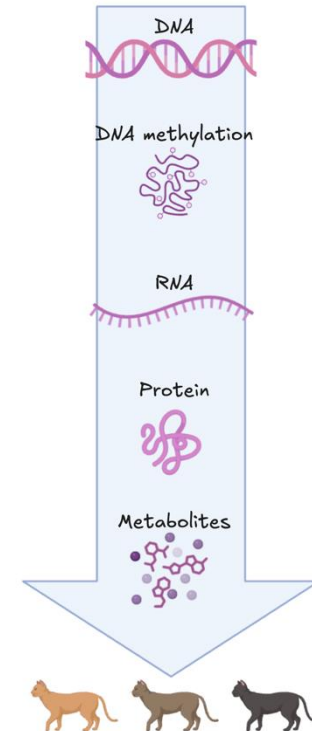
A quick break!

Module 2: Designing robust omics studies

Recap

In Module 1 we explored:

- How the scientific question determines which of the five molecular layers to measure
- Considerations across the omics study design workflow



Learning objectives

- Select an appropriate platform to capture the biological signal of interest and describe the type of data it produces.
- Evaluate a study design in terms of accuracy and interpretability, power and cost, and generalisability.

Key aspects of study design

Aspect	Design question to consider
2.1 Accuracy and interpretability	• Will the measurements accurately represent the biological feature of interest? • Can the results be interpreted with confidence?
2.2 Power and cost	• Can the study detect the biological effect of interest within the available budget?
2.3 Generalisability	• Are the findings likely to apply beyond the study cohort or experimental system?

Module 2.1.1: Accuracy and interpretability

2.1.1 Learning objectives

- Explain why platform, library-prep, and acquisition decisions must be made before data collection and cannot be revisited afterwards
- Distinguish between sequencing-based platforms (reads) from signal-based platforms (intensities)
- Evaluate a library-preparation and read-length choice against the biological features it can and cannot resolve
- Compare label-free vs labelled quantification, and DDA vs DIA, in terms of the missing-value patterns each produces

Choosing a measurement platform

1. Sequencing-based

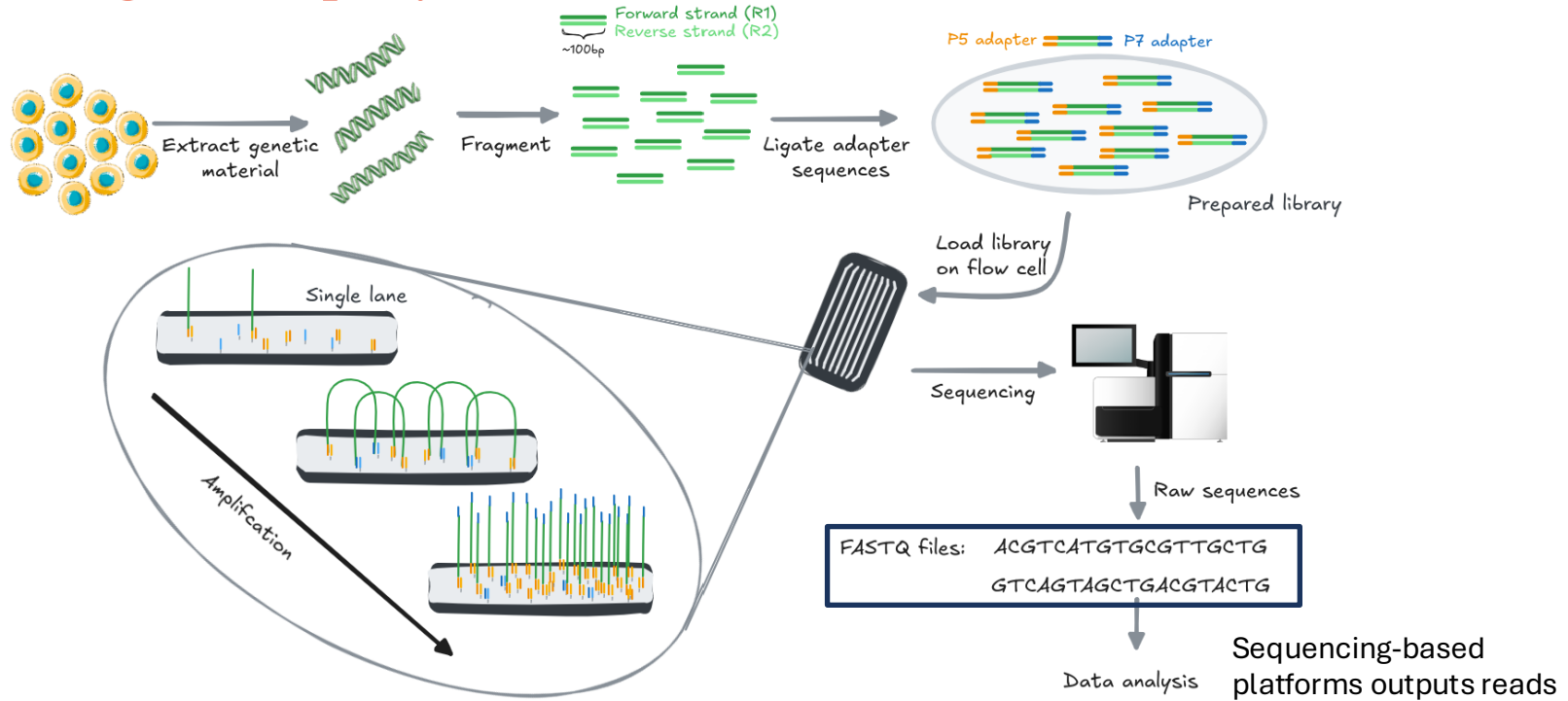
	MOLECULE	PLATFORMS	DATA TYPES	ANALYSES	WHEN TO USE
Genomics	DNA 	Whole genome sequencing whole exome sequencing	Sequence reads	Variant calling, copy number alterations, structural variants	WGS: coding and non-coding genome WES: coding only
Epigenomics	DNA methylation 	ATAC-seq, ChIP-seq, bisulfite-seq	Sequence reads	Chromatin accessibility, histone modification, DNA methylation	ATAC-seq: open chromatin ChIP-seq: protein-DNA binding
Transcriptomics	RNA 	Whole transcriptome sequencing, bulk RNA-seq, single-cell and spatial transcriptomics	Sequence reads	Gene expression, cell-type composition, isoform and splicing	WTS: transcriptome assembly Bulk RNA-seq: tissue-level gene expression
Proteomics	Protein 	Mass spectrometry, Olink	Signal intensities, proximity extension assay counts	Protein abundances, modifications, interactions	MS: unbiased unknown protein discovery Olink: targeted panel
Metabolomics	Metabolites 	Mass spectrometry, nuclear magnetic resonance spectroscopy	Signal intensities	Metabolite abundances and profiles, pathway activity	MS untargeted: unknown metabolite NMR: absolute quantification

2. Signal-based

Phenomics

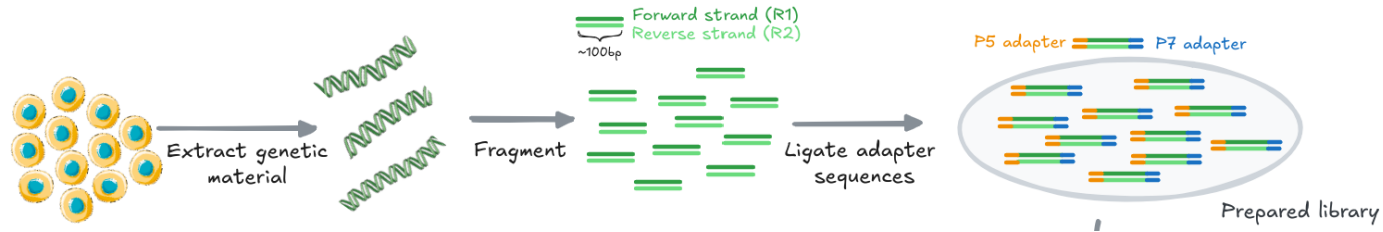


Sequencing-based platforms

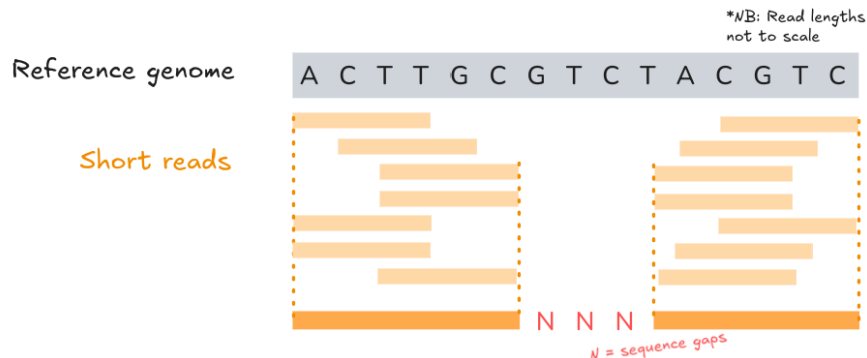


Library preparation

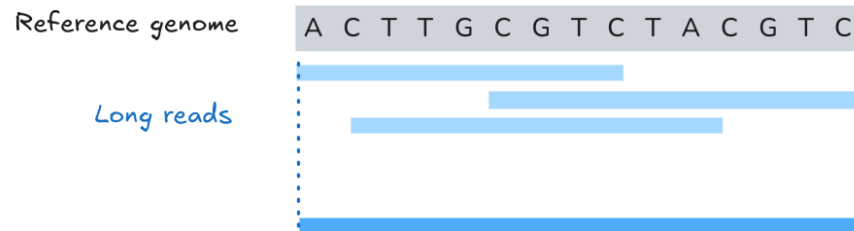
- Library preparation converts extracted nucleic acid into a form the sequencer can process
- Enrichment captures the target molecule (DNA, exomes, RNA, polyadenalated RNA etc.)
- Multiplexing: multiple samples are barcoded and sequenced together



Read length



Short-read sequencing

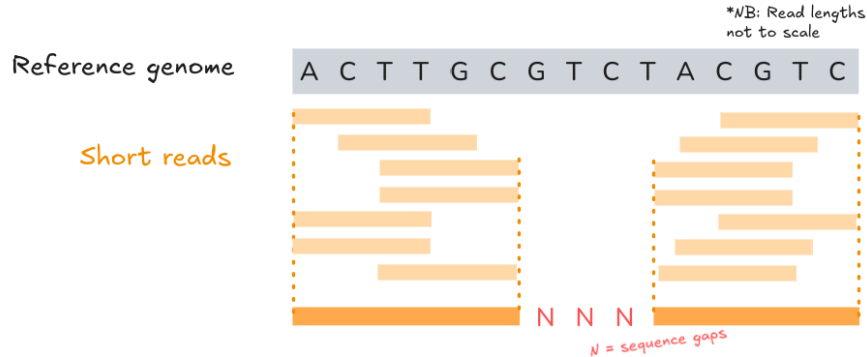


Long-read sequencing

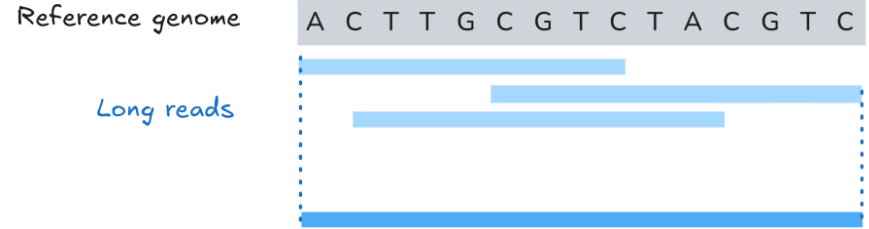
Property	Short-read sequencing	Long-read sequencing
Base-level accuracy	Higher	Lower
Cost per-base	Cheaper	More expensive
Resolves structural rearrangements and repetitive regions	Poorly	Well

Read length

Short-read sequencing



Long-read sequencing

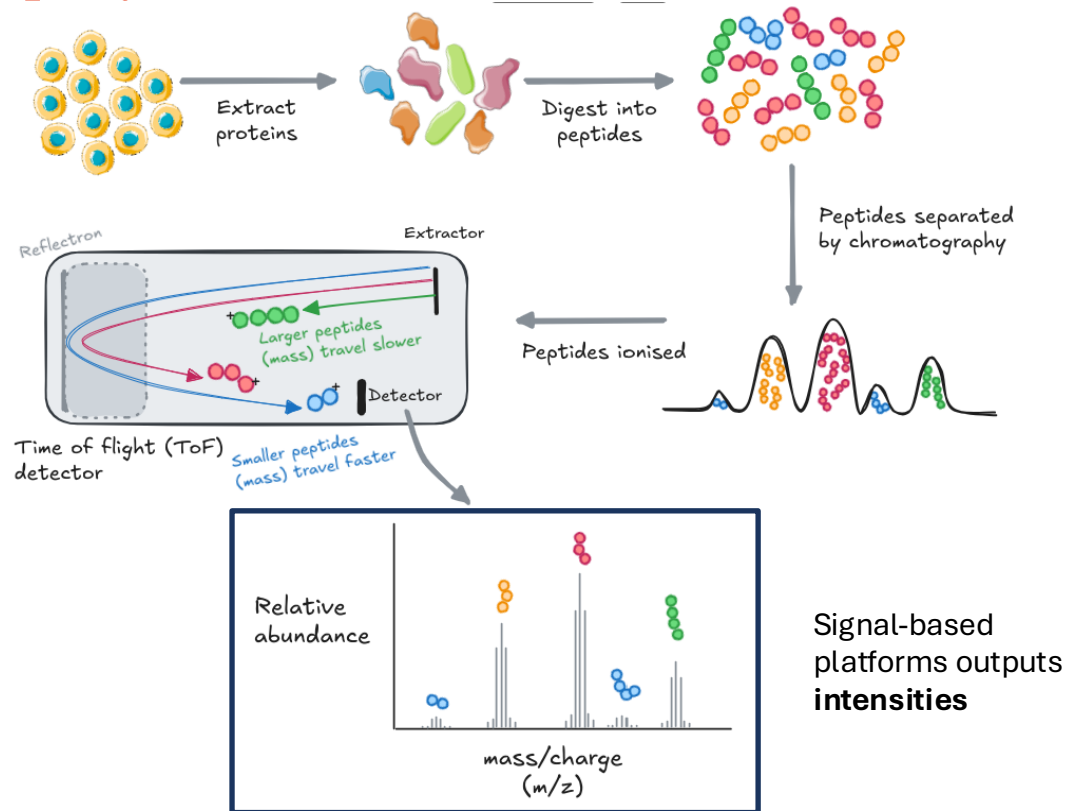


Decide whether short reads, long reads, (or both) are most suitable for resolving each of the goals:

1. Identifying a rare single nucleotide variant in a gene you already know to look at.
2. Assembling a small bacterial genome with numerous structural rearrangements.
3. Assembling a large vertebrate genome from scratch (de novo).



Signal-based platforms



Label-free and labelled quantification



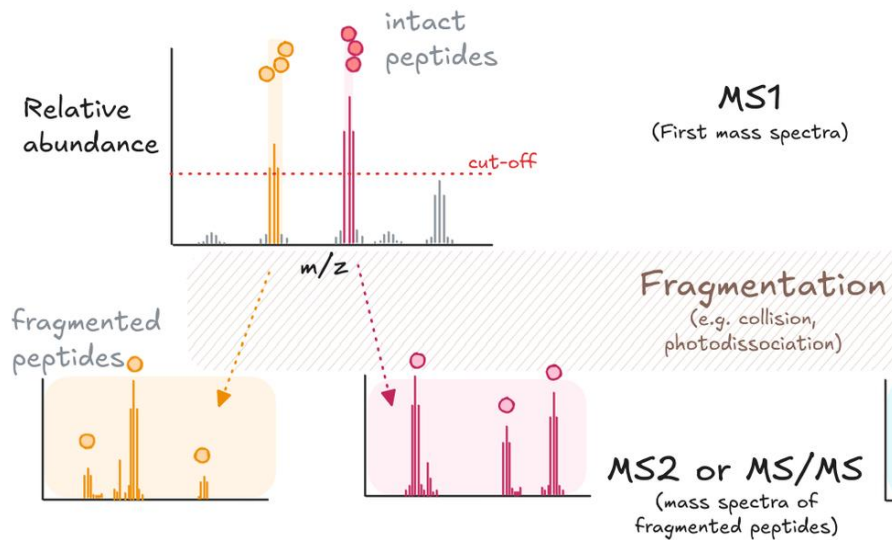
Property	Label-free	Labelled (tandem mass tag)
Samples per run	• One sample • No limit	• Multiple samples pooled together (multiplexed) • Limited samples (6~18) based on labelling scheme
Instrument run time	Faster (one sample to process)	Slower (multiple samples to process)
Variation in measurement	Can vary across runs	No run-to-run variation
Best suited for	Large sample numbers	Studies requiring high consistency across samples

Select one of the scenarios. Decide whether label-free or labelled quantification is the better fit:

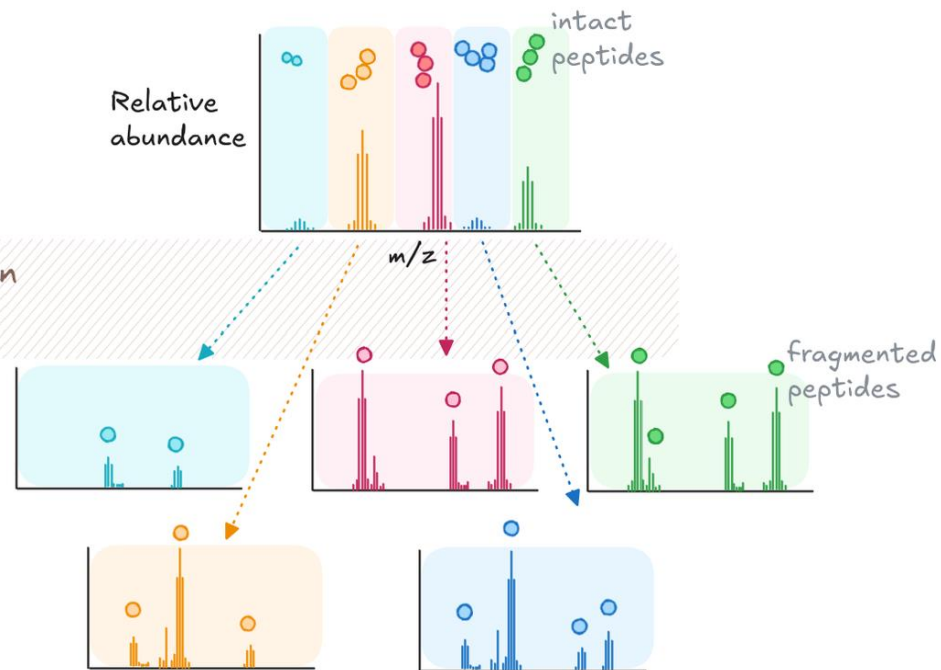
1. Discovering candidate biomarkers of early-stage disease across 150 patients
2. Resolving subtle abundance changes across 6 replicate cell cultures before and after drug treatment
3. Profiling 15 samples from a developmental time course

Acquisition mode

Data Dependent Acquisition

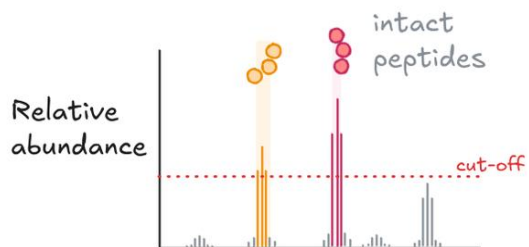


Data Independent Acquisition



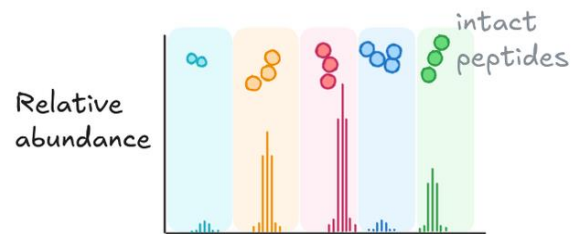
Acquisition mode

Data Dependent Acquisition



MS1
(First mass spectra)

Data Independent Acquisition



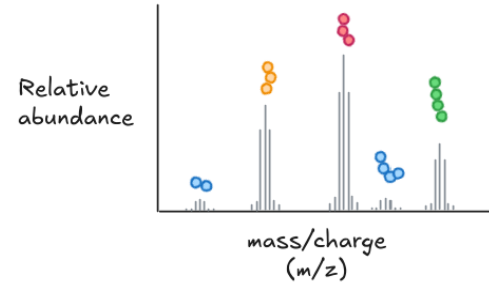
Property	DDA	DIA
Ion selection	Selects most abundant ions	Fragments all ions across predefined m/z windows
Coverage	Biased for high-abundance ions	Consistent across windows
Missingness	Consistently misses low-abundance ions	Reduced
Analysis complexity	Standard	Complex, requires specialised software for interpretation
Best suited for	Targeted discovery of abundant ions	Consistent and reproducible coverage across runs

Key decisions



Sequencing-based platforms:

- **Library preparation:** Will it enrich the right molecule?
- **Read length:** Will you have the correct coverage to resolve region you are studying?



Signal-based platforms:

- **Quantification:** Will it have the same consistency across runs?
- **Data acquisition:** Will it resolve the subtle abundance differences you're looking for?

Activity: platform audit



Improve your research question

Next to your research outline on the wall, add another sticky note, indicating whether you are using either a:

- **Sequencing-based platform, or**
- **Signal-based platform**
- or both

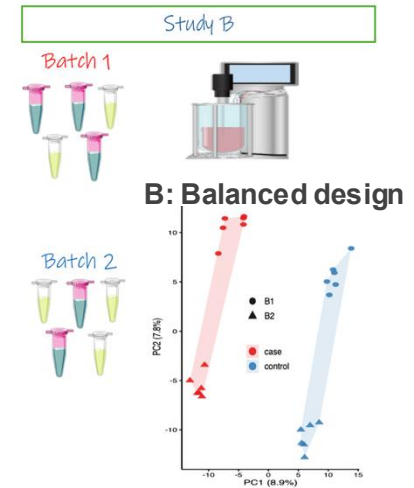
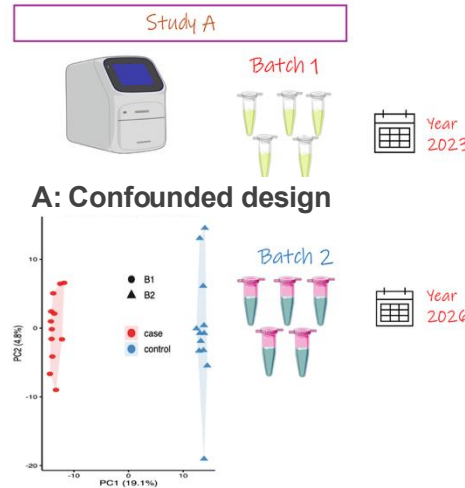
Module 2.1.2: Confounding and batch

2.1.2 Learning objectives

- Interpret an example study design to identify when a group difference may be confounded rather than biological
- Explain why no downstream correction can recover confounded variables
- Compare why blocking corrects for batching in named factors, and randomisation for unnamed ones

Confounding

- Confounding occurs when variables are associated with the condition/exposure and could also affect the omics measurements.
- Example variables include sex, age, disease state, medication use, tissue composition, sample handling
- **A confounder that was neither controlled nor recorded cannot be modelled during analysis.** If a variable may influence the outcome, measure it at the time of collection.



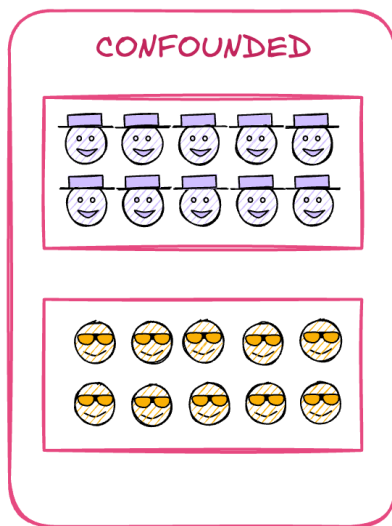
Key Terms

Term	Examples
Phenotype/condition	Disease status, crop yield, treatment group
Omics feature (predictor)	Mutation, protein abundance, metabolite concentration
Metadata variable	Age, sex, tissue type, processing batch

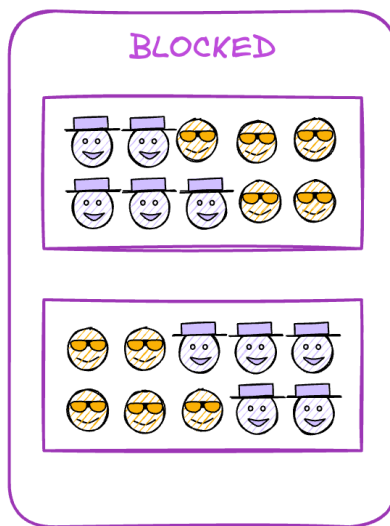
Key terms for identifying confounders

Term	Examples
Phenotype or condition	Disease status, crop yield, treatment group
Omics feature (predictor)	Mutation, protein abundance, metabolite concentration
Metadata variable	Age, sex, tissue type, processing batch

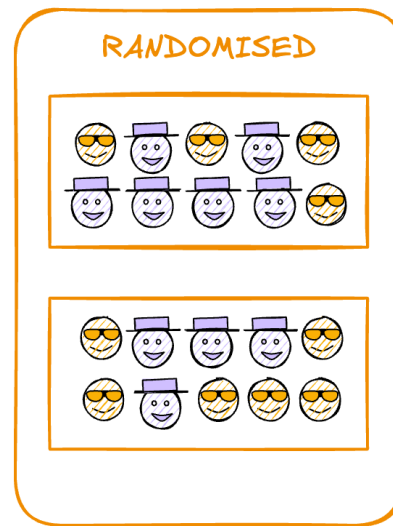
Design approaches to mitigate confounding and batch effects



Batch cannot be distinguished from the biology

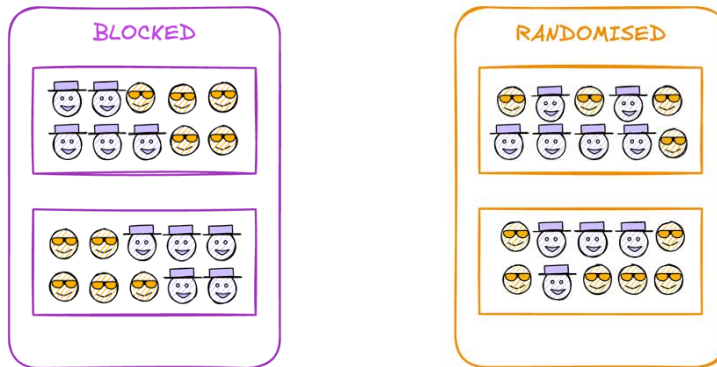


Batch estimatable removable from within-batch contrasts



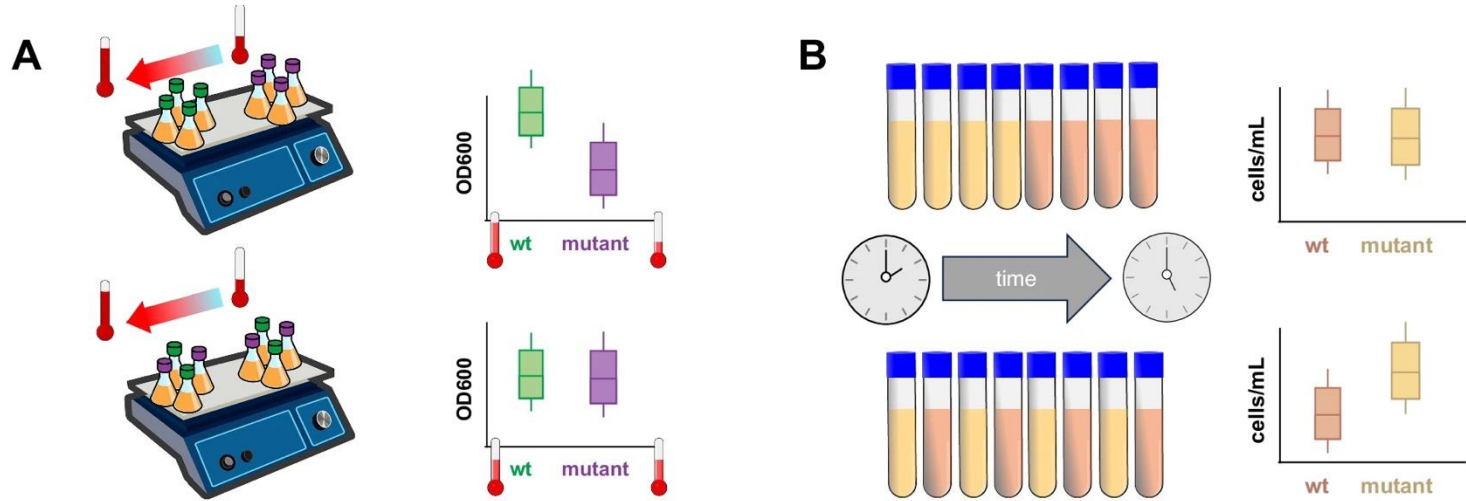
Chance balance reduces systematic bias

Design approaches to mitigate confounding and batch effects



Property	Blocking	Randomisation
Manages sources of variation that...	Can be planned for	Cannot be managed in advance
Sample allocation	Deliberatively allocated across factors	Processed in a mixed order
Avoids...	Confounding with the comparison of interest	Aligning with the comparison of interest

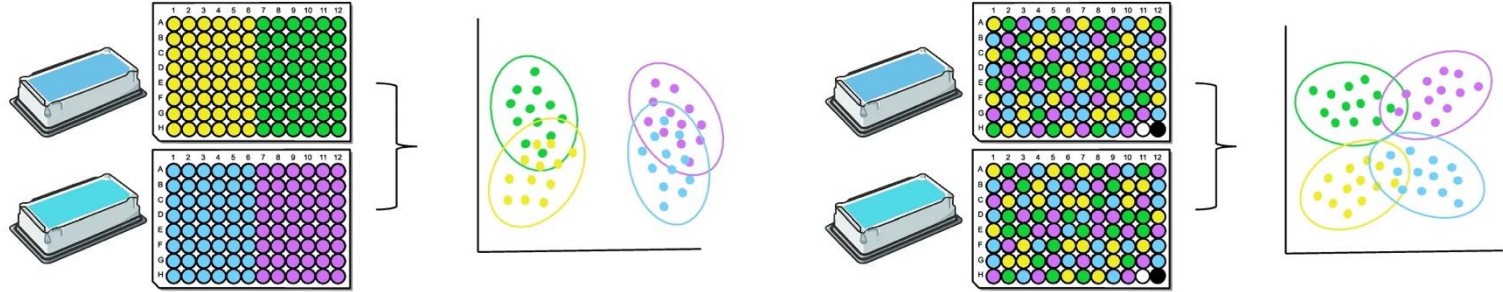
Randomisation to avoid unknown variation



Wagner & Kleiner. *Nature Communications* 16, 7263 (2025)

Blocking to manage known sources of variation

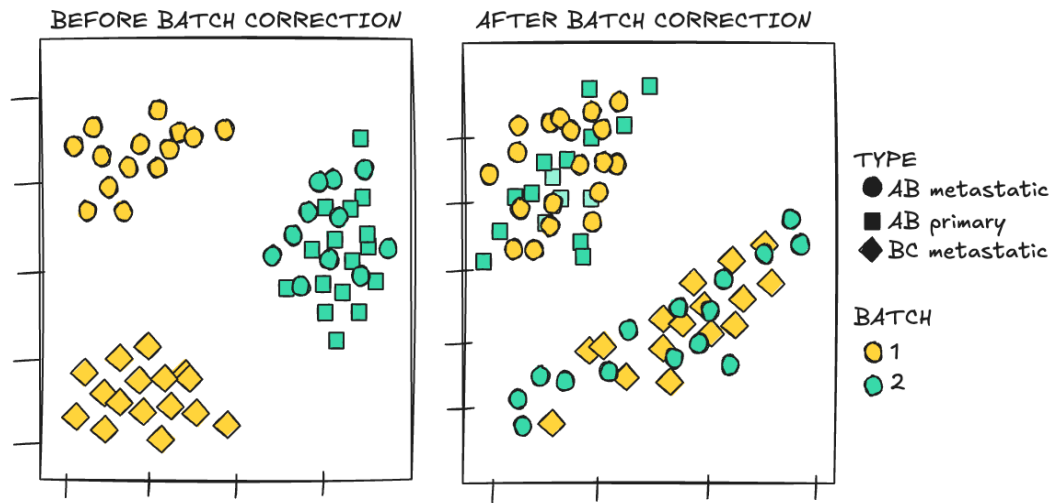
C



Wagner & Kleiner. *Nature Communications* 16, 7263 (2025)

Design, correction, and interpretation

- Confounding and batch effects cannot always be prevented at the design stage
- Inheriting datasets, reusing public data, working with rare samples, working within budget constraints
- Batch effects can be corrected in the analysis stage only if the design information was recorded



[Zhu et al. Genome Medicine 9, 108 \(2017\)](#)

Module 2.1.3: Measurement and metadata

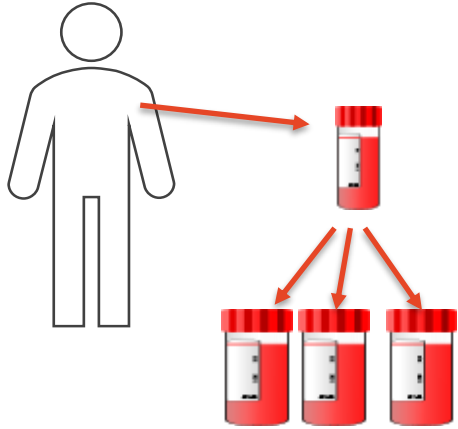
2.1.3 Learning objectives

- Determine whether technical replication is warranted for a given platform and study design.
- Explain how a shared reference sample enables batch effect estimation and correction.
- Identify the biological, technical, and contextual variables that should be recorded for a given study.
- Articulate why unrecorded variables cannot be recovered and how this constrains interpretation.

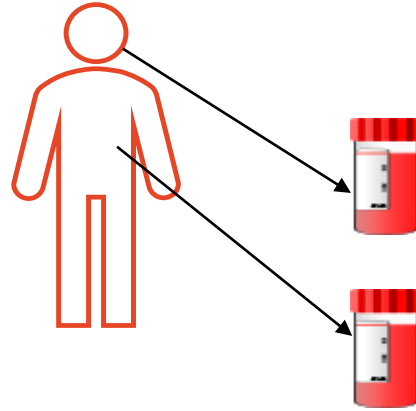
Measurement reliability and metadata

A well-designed experiment still requires **reliable measurements** and **sufficient context**

Technical replicates and repeated measurements



A **technical replicate** measures the same sample more than once.



A **Repeated sample** measures the same biological unit again at a different site

Technical replicates

	Technical	Biological
Assesses measurement consistency	Yes	No
Supports quality control	Yes	No
Informs measurement consistency	Yes	No
Contributes to statistical power and sample size	No	Yes

Use technical replicates when:

- Measurement noise is high
- Measurements span multiple batches or runs

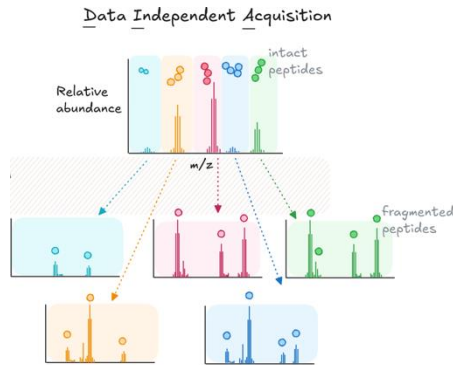
Shared reference samples can remove noise

- When the same sample is measured more than once, differences are often due to measurement (technical) variability rather than biology
- A **shared reference sample** is typically a pooled aliquot derived from all study samples
- The shared reference sample is measured across all runs to assess the drift across batches and can be used to subtract the differences across these



Acquisition mode and technical replication

Recall that **data-independent acquisition (DIA)** fragments ions across fixed m/z windows, rather than selecting only the most abundant ions. This gives more consistent sampling **across runs**, but the instrument signal can still drift over time.



A researcher is planning a DIA proteomics study with 80 plasma samples measured across four batches over two weeks. They want to track whether the instrument signal changes during acquisition.

Which design choice would best help them separate instrument drift from biological differences between samples?

- A. Technical replicates are not needed because DIA is a highly reproducible platform
- B. Measure a subset of study samples twice and treat them as extra biological replicates
- C. Inject the same pooled reference sample at regular intervals across all four batches
- D. Wait until analysis, then add technical replicates if a PCA plot shows batch structure

Metadata gives measurements context

- Metadata is a structured record of the biological and technical variables associated with each sample.
- **Within a study:**
 - Allows observed patterns to be investigated
 - Allows resolution and correction of confounding or batch effects
 - Supports interpretation of results
- **Beyond a study:**
 - Allows reuse of data in future studies (secondary analyses, meta-analyses, etc.)
 - Correct and complete metadata is very important for reproducibility (FAIR principles)

sample_id	patient_id	condition	tissue_site	sex	age	age_group
S001	P01	Disease	blood	M	68	65+
S002	P01	Disease	biopsy	M	68	65+
S003	P01	Disease	saliva	M	68	65+
S004	P02	Disease	blood	F	72	65+
S005	P02	Disease	biopsy	F	72	65+

A minimal metadata checklist

- The purpose of this checklist is to ensure all variables capable of explaining structure in the data is recorded.
- Variables are grouped by the phase at which they should be captured.

Stage	Example variables
Collection	Sample type, condition, biological unit, age, sex, collection time
Handling	Preservation method, processing delay, freeze–thaw cycles
Processing	Extraction batch, reagent lot, input quantity, quality metric
Measurement	Instrument, acquisition batch, run identifier

Activity: metadata audit



Group activity: metadata audit

Each module 2.1.3 scenario on your desk describes a real study design problem. You are not expected to know the answer, the goal is to find it.

1. Read the scenario.
2. Open a web browser and search engine of your choice.
3. Use the provided search term as a starting point.
4. Write a short answer to the **find** question on each scenario page in your own words.



Scenario: Whole genome sequencing

A whole-genome sequencing study compares tumour tissue from two patient groups. Samples from Group A were collected prospectively and stored as fresh frozen; samples from Group B were retrieved from a hospital biobank where tissue had been archived as FFPE blocks. Preservation method was not recorded in the study metadata.

Search: FFPE DNA sequencing artefacts

Find: What type of chemical damage does FFPE fixation introduce into DNA, how does it appear in sequencing data, and why does it create a problem when FFPE and fresh -frozen samples are analysed together without a recorded preservation variable?



Scenario: ChIP-seq

A ChIP-seq experiment targets a histone modification associated with active transcription. Libraries were prepared for the immunoprecipitated (IP) fraction only; no input control library was generated. After sequencing, peaks are called and used to identify regions of differential modification between two conditions.

Search: ChIP-seq input control purpose

Find: What does an input control represent in a ChIP-seq experiment, what sources of signal does it capture, and what cannot be determined about a called peak when no input control is available?



Scenario: Gene expression

A bulk RNA-seq study compares gene expression in peripheral blood between two disease subtypes. Samples were collected during routine clinical visits: subtype A patients were typically seen in morning clinics and subtype B patients in afternoon clinics. Collection time was not recorded in the study metadata.

Search: circadian gene expression peripheral blood

Find: What proportion of the expressed genome shows circadian rhythmicity in peripheral blood, and what does this imply for a study where collection time is systematically different between comparison groups?



Scenario: Label-free proteomics

A label-free quantitative proteomics study compares plasma samples from two clinical sites. Sample preparation, including depletion of high-abundance proteins, reduction, alkylation, and tryptic digestion, was performed by different operators at each site. Operator identity was not recorded in the metadata.

Search: pre-analytical variation label-free proteomics sample preparation

Find: Which steps in proteomics sample preparation are documented sources of inter-operator or inter-laboratory variation, and what is the magnitude of that variation relative to typical biological effect sizes?



Scenario: Untargeted metabolomics

An untargeted metabolomics study analyses plasma samples from 120 participants using reverse-phase LC-MS. Samples were acquired in a single continuous run lasting approximately 14 hours. No pooled QC samples were included in the acquisition sequence.

Search: pooled QC samples metabolomics instrument drift

Find: What is the purpose of pooled QC samples in a metabolomics acquisition, what signal do they capture across a run, and what cannot be determined or corrected when they are absent?



Improve your research question

Next to your existing sticky notes, add sticky notes for metadata that you have or plan to collect.

Minimal checklist:

Stage	Example variables
Collection	Sample type, condition, biological unit, age, sex, collection time
Handling	Preservation method, processing delay, freeze-thaw cycles
Processing	Extraction batch, reagent lot, input quantity, quality metric
Measurement	Instrument, acquisition batch, run identifier

Module 2.1 summary: accuracy and interpretability

Area	Summary
Platform selection	Measure the right molecule at the required resolution
Confounding and batch	Distribute groups across other sources of variation
Measurement reliability	Use technical controls to identify measurement variation
Metadata	Record variables that could affect the measurements

Module 2.2: Power and cost

Key aspects of study design

Aspect	Design question to consider
2.1 Accuracy and interpretability	• Will the measurements accurately represent the biological feature of interest? • Can the results be interpreted with confidence?
2.2 Power and cost	• Can the study detect the biological effect of interest within the available budget?
2.3 Generalisability	• Are the findings likely to apply beyond the study cohort or experimental system?

Module 2.2.1: Sample size

2.2.1 Learning objectives

- Explain why statistical power is necessary to interpret analysis results
- Identify design-stage strategies to ensure a study is well-powered (replication, platform-appropriate minimum sampling, detectable effect sizes)
- Describe analysis-stage strategies for validating power
- Explain why omics requires its own statistical methods, rather than classical power calculations

Sample size (n)

A study's sample size (n) is the number of independent biological units being compared.

Independent biological unit examples:

- Independent patients
- Independently treated animals
- Independent cultures
- Independent environmental units

Non-biological units:

- Repeated measurements
- Technical replicates
- Multiple cells from one donor
- Subsamples from one location

Inflated sample sizes

When subsamples are treated as individual units, n is inflated and can result in pseudoreplication. This leads to misleading p-values.

	Sampling	Examples
Subsampling	Takes many measurements from one biological unit	• Thousands of cells from one donor • Multiple vials from one water body
Pooling	Merges material from several biological units before measurement	• Combining soil samples from multiple field sites to represent an average measurement

Valid replication or pseudoreplication?

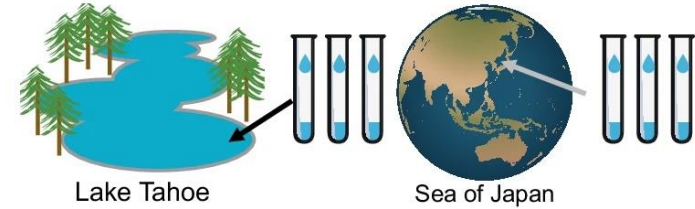
A team wants to know whether microbial communities differ between freshwater and marine environments. Two sampling designs (A and B) are proposed. Both produce six samples and cost the same to sequence.

Q1: What counts as an independent biological replicate for comparing freshwater and marine environments? Explain why.

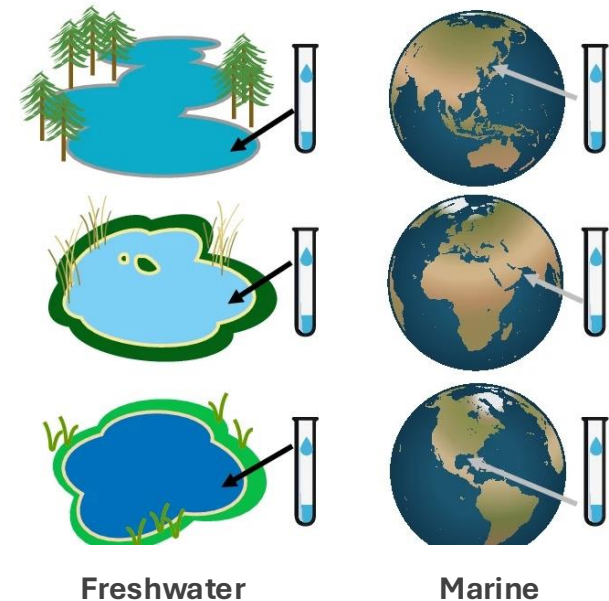
Q2: How many independent biological replicates (n) are there per in each design?

Q3: In which design would treating all three samples per environment as independent replicates lead to pseudoreplication? Explain why.

A



B



Statistical power in omics

Omics budgets are finite.

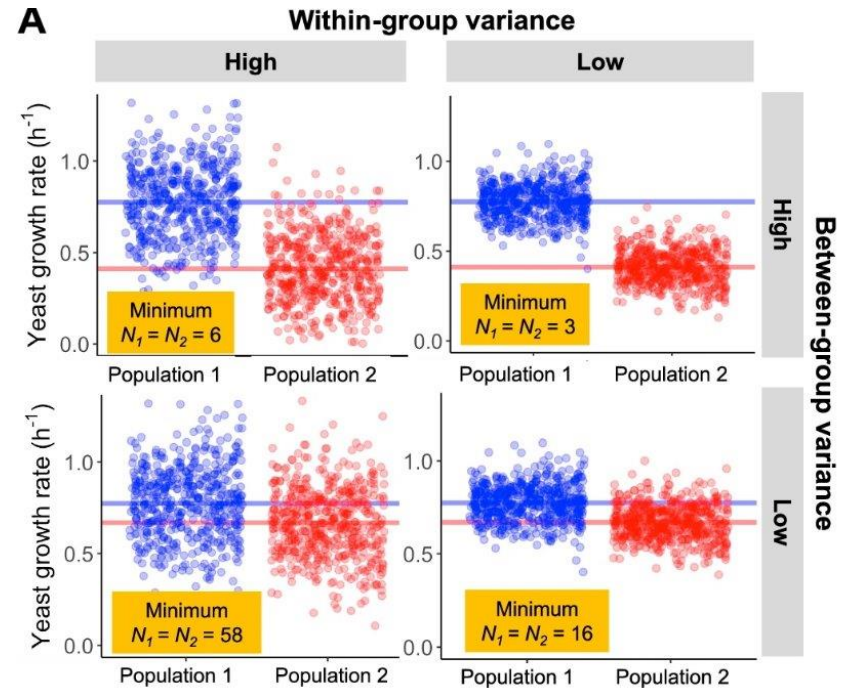
Can we measure enough independent samples, at sufficient depth, to detect the biological difference that matters?

What determines statistical power?

- Number of independent samples
- Expected effect size
- Biological variability
- Significance threshold and multiple testing

Practical constraints for most omics studies

- Available budget
- Sample availability



Estimating the sample size

A power calculation is a formula that **estimates the sample size needed to detect an effect of a given magnitude, with a given probability, at a given significance threshold**. It requires three inputs:

Input	Description
Effect size	How large a difference is being detected
Biological variability	How much samples differ from each other for reasons unrelated to the effect
Multiple-testing burden	How many features are being tested and corrected simultaneously

Effect size

Effect size: How large the biological difference is between the groups or conditions being compared.

Examples:

- A twofold difference in protein abundance between treatment and control.
- A variant is associated with 1.5 times higher odds of disease (**OR = 1.5**).

Estimating the effect size of omics:

- **Domain knowledge:** define a meaningful effect
- **Comparable datasets:** estimate variability in a similar system
- **Pilot data:** simulate performance across possible sample sizes
- Use evidence from the relevant tissue, platform and comparison.

Biological variability in omics

Biological variability: Differences among independent biological samples that are unrelated to the condition being studied.

Lower variability	Higher variability
Groups are easier to distinguish	Groups overlap more
Fewer samples may be needed	More samples are usually needed

Examples: Genetic background, age, environment, and individual biological differences.

Estimate variability using pilot data or comparable studies.

Minimum sample sizes (n) across platforms

Published lower bounds you come across in the literature (should) reflect the conditions of the studies that generated them (the specific tissue, organism, platform configuration, and effect size). A number that was adequate in one system may be inadequate in another with higher biological variability or a more subtle effect.

These ranges are a starting point for the sample size conversation, not a universal threshold.

Layer	Platform(s)	Indicative range
Genome	WGS, WES	100-1000 donors for association studies. Rare variant studies require more.
Epigenome	ChIP-seq, ATAC-seq, WGBS	≥3 per condition for differential analysis. More when effect sizes are small
Transcriptome	Bulk RNA-seq	≥6 per groups for moderate-large effects. ≥12 when studying smaller effects.
Transcriptome	scRNA-seq	≥6 donors per group to start. Cell number per donor is a secondary consideration.
Proteome	Label-free MS	No universal minimum. Effect n per protein is lower than total samples run
Metabolome	LC-MS, GC-MS	5-10 per group for

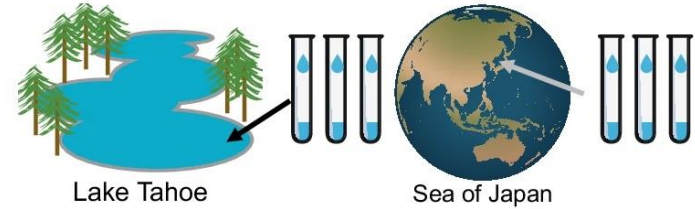


What does $n = 3$ mean for power?

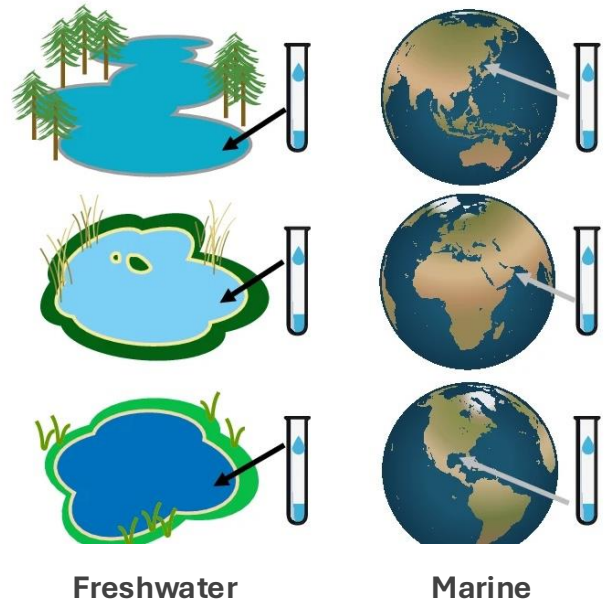
Design B has $n = 3$ water bodies per environment. A 16S amplicon sequencing analysis will test thousands of microbial features simultaneously, with false discovery rate correction applied.

1. Is $n = 3$ per environment likely to be sufficient for this analysis? What additional information would be needed to answer this properly?
2. If the study fails to detect any significant differences after FDR correction (recall from Module 1), what are the possible explanations?

A



B



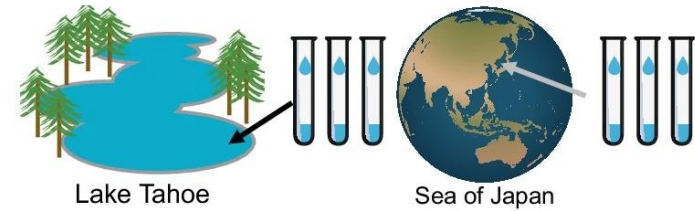


How would you determine the right n ?

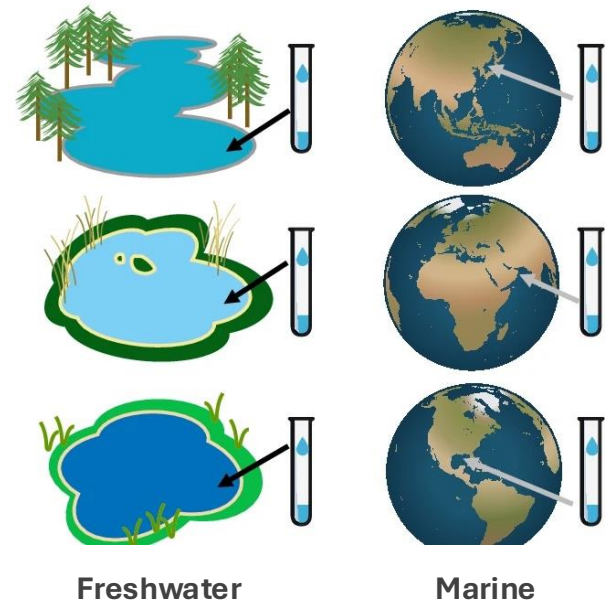
The fieldwork budget allows for a maximum of 5 water bodies per environment. Before committing to this design, the team wants to know whether $n = 5$ is sufficient.

1. What information is needed to run a power calculation for this study?
2. Where would that information come from?
3. If the estimate suggests $n = 5$ is underpowered, what are the options?

A



B



Estimating power for omics studies is challenging!

Key takeaways:

- Ensure you have sufficient sample size (n) and not pseudoreplicates
- Estimate the expected effect sizes and biological variability in your studies to determine the adequate sample size required
- Account for platform-specific guidelines
- **Use evidence from existing studies that match your study organism and comparisons**

Module 2.2.2: Design decision costs

2.2.2 Learning objectives

- Explain why depth and replication serve different analytical functions and cannot substitute for each other
- Describe why returns on additional depth diminish once the detection floor is met
- Identify how multi-omics designs compound budget constraints
- Apply a decision sequence for allocating a finite budget across depth, replication, and platform scope

Budget trade-offs

Beyond sufficient replicates, omics measurements requires a strong enough signal to be detected (depth). Three design choice therefore compete for the same budget:

1. How many biological samples to collect
- 2. How deeply each sample is measured**
- 3. Whether to measure one or several molecular layers**

Measurement depth across omics platforms

Every platform has a detection floor – **the depth below which a feature of interest is inconsistently sampled or missed entirely across samples.**

Layer	Platform(s)	Depth
Genome	WGS, WES	Coverage – the number of times each base in the genome is sequenced
Epigenome	ChIP-seq, ATAC-seq	The number of reads mapping to each genomics region. Must exceed background signal.
Epigenome	WGBS	The number of times each cytosine position is sequenced after bisulfite conversion
Transcriptome	Bulk RNA-seq	Total reads per sample. Determines detection of low-expression genes
Transcriptome	scRNA-seq	Reads per cell (per-cell completeness) and cells per donor (resolution, not power)
Proteome	LC-MS/MS	Peptides sampled. Determined by acquisition mode (DIA/DDA), gradient length and fractionation
Metabolome	LC-MS/GC-MS	No universal minimum. Effect n per protein is lower than total samples run

Replicates do not contribute to measurement depth

- **Depth:** determines whether the feature is detectable/detected
- **Replication:** provides statistical power for detecting differences between groups
- Even if an experiment has sufficient replication and sample size, **if depth is insufficient (lower than the detection floor) the feature of interest will still be missed.**

Considerations for choosing depth

1. Once measurement depth exceeds what is needed to consistently detect the features of interest, **additional depth beyond the detection floor diminish fast.**
2. Aim to **match the measurement depth across runs and comparison groups**, otherwise it can be a confounding variable; a special case of batch effect confounding
3. **The detection floor is set by the biology that informs your research question**, not the platform

Omics applications requiring high depth

Layer	Example	Fix
Genome	Rare somatic variants (5% of cells)	Increase coverage to 200-1000X
Transcriptome	Low-abundance or regulatory transcript	Increase reads per sample
Proteome	Low-abundance or co-eluting protein	Switch to DIA, longer gradient lengths, or fractionation
Metabolome	A trace metabolite near detection limit	Switch to targeted acquisition

Considerations for choosing depth

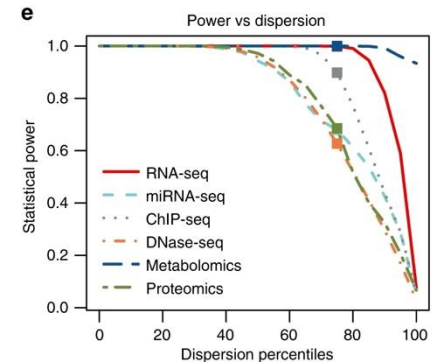
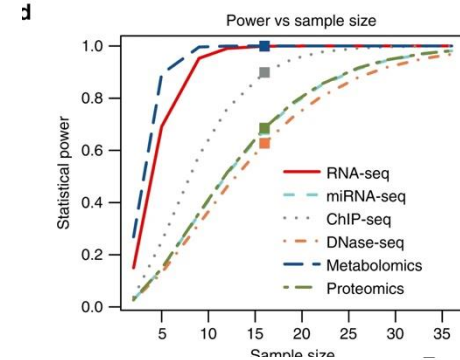
1. Define the features the question depends on
2. Determine the minimum depth needed to detect those features consistently
3. Set depth at that minimum. Do not exceed it without a specific analytical reason
4. Determine the minimum biological sample number using a power calculation
5. Account for controls and expected attrition before finalising sample number
6. If the budget cannot support both minimum depth and minimum sample number, **adjust scope**

Allocating budget across multiple molecular layers

Multimomics: Multiple molecular layers measured from the same samples to improve understanding of the underlying biology

Design considerations for multimomics experiments:

- Measuring multiple layers increases the cost
- The minimum sample number is determined by the most demanding platform
- Prefer a single adequately powered platform study over an underpowered multi-platform study





Group activity: allocating a constrained budget

1. For each design, is depth set at or above the detection floor for the features the question depends on?
2. What does increasing depth from 30 to 60 million reads in Design A provide for this question?
3. What does Design B gain relative to Design A, and what does it lose?
4. What question can Design C answer that neither A nor B can?
5. Which design is most appropriate for this question, and why?



Group activity: allocating a constrained budget (p 1/3)

A team is designing a study to compare energy metabolism in liver tissue between animals exposed to a metabolic stressor and untreated controls. Their hypothesis, grounded in published literature, is that the **stressor shifts hepatic substrate use from fatty acid oxidation toward glucose oxidation.**

As Module 1.1 established, the transcriptome captures which genes are active, while the metabolome captures the biochemical state of the cell at the moment of sampling. Both layers are informative for this question:

- **Transcriptomics would show whether genes encoding fatty acid oxidation enzymes and glucose metabolism enzymes are differentially expressed**
- **Metabolomics would show whether the predicted metabolic shift is occurring at the level of actual metabolite concentrations.**



Group activity: allocating a constrained budget (p 2/3)

Their budget supports one of three designs:

Design	Platform	Depth	<i>n</i> per group
A	Bulk RNA-seq only	60 million reads per sample	6
B	Bulk RNA-seq only	30 million reads per sample	12
C	• Bulk RNA-seq, and • Targeted metabolomics (80 pre-specified metabolites covering fatty acid oxidation, TCA cycle, and glycolysis)	• 30 million reads per sample • Standard targeted acquisition	6



Group activity: allocating a constrained budget (p 3/3)

Published RNA-seq benchmarking in mouse liver shows:

- At 30 million reads, approximately 85-90% of expressed genes are detected
- This proportion does not increase substantially at 60 million reads.
- The detection floor for the transcripts relevant to this question (genes encoding metabolic enzymes) are expressed at moderate-to-high levels in the liver
- The detection floor for the relevant transcripts is below 30 million reads per sample
- For the targeted metabolomics panel, all 80 analytes are expected to be consistently detected above instrument detection limits under standard acquisition conditions

Module 2.3: Generalisability

Generalisability: who and what the findings apply to

Generalisability asks whether findings extend beyond the samples and conditions studied.

1. Population: Which people, organisms or environments do the findings describe?
2. Biological scope: Which tissue, molecular layer and time point were measured?

A trustworthy finding can still have a narrow scope.

Internal and external validity: two different questions

- **Internal validity:** Is the comparison within this study trustworthy?
- **External validity:** Do the findings apply beyond this study?
- Example: A well-controlled proteomics study in mouse may provide a reliable comparison, but its findings may not extend to humans.

What limits generalisability?

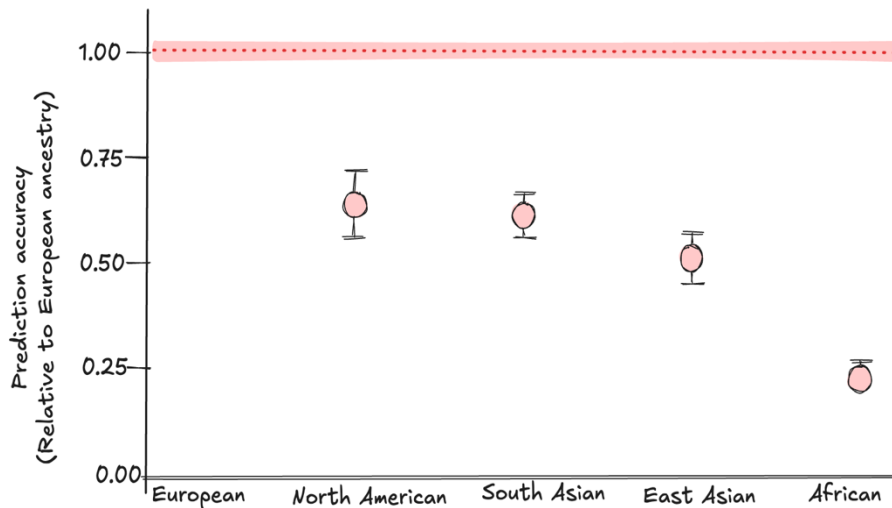
- **Who was sampled:** Sex, age, species, strain or variety.
- **Where and when:** Recruitment site, environment, season or collection time.
- **Biological context:** Disease stage, treatment or experimental conditions.
- **What was measured:** Tissue, molecular layer and platform coverage.

Examples:

- Findings from **adults** may not apply to **children**.
- Higher immune-related protein abundance in a tumour sample may reflect more immune cells, rather than upregulation of those protein by tumour cells.

Case study: polygenic risk scores and ancestry

- **Polygenic risk scores** combine information from many genetic variants to predict a trait or disease risk.
- Scores developed mainly from European-ancestry data performed less accurately in other ancestry groups.
- **Who we recruit affects who our findings apply to.**



Source: Martin et al., Nature Genetics (2019).

Power and generalisability: a design trade-off

A narrower cohort

- May reduce biological variability and improve power.
- Supports conclusions about a more restricted population.

A broader cohort

- Includes more of the population we want to understand.
- May require more samples and careful management of confounding and batches.

Under a fixed budget, choose the scope that best serves the research question.

More samples from the same narrow population do not automatically broaden applicability.

How can we strengthen generalisability?

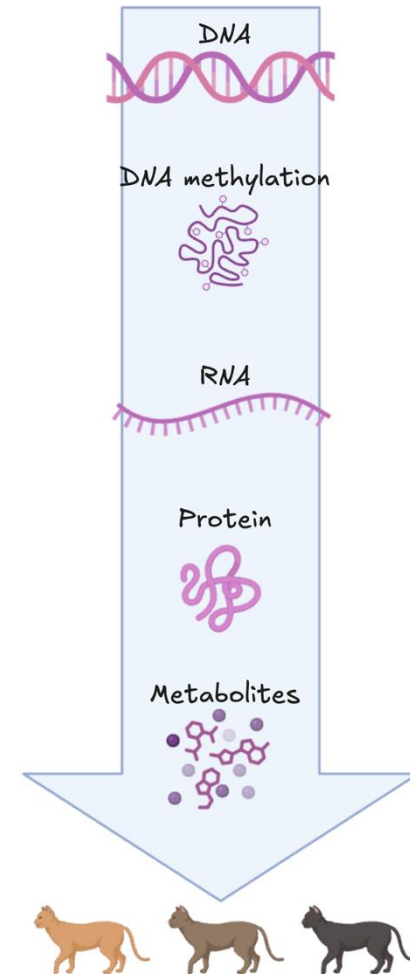
- **State the scope:** Name the population, tissue, molecular layer and collection conditions.
- **Broaden the design:** Include relevant sites, populations or conditions while preserving fair comparisons.
- **Test independent data:** Assess whether findings hold in suitable public datasets or a new cohort.

Module 3: Diagnosing design problems

Recap

In Module 1 we explored:

- How the scientific question determines which of the five molecular layers to measure
- Considerations across the omics study design workflow



Recap

In Module 2 we explored:

- How platform selection determines what biological signal can be captured
- How to evaluate study designs for accuracy and interpretability, power and cost, and generalisability

Omics-calculator demo

Activity: Diagnose the case study design

Diagnose the case study design



<https://sih.tools/bio-omicscalculator>

Apply your learnings from this workshop to the provided case studies. Pick a case study: A, B, or C and use the omics calculator to explore the case study's samplesheet. Answer the questions about study design.

1. What is the main comparison, and does the omics type answer the research question?
2. Are there variables that differ between groups that could affect the outcome?
3. Could batch or technical effects influence the results, and were they captured?
4. Could these findings apply to populations outside this study?
5. How would you improve the design?



Module 3: case study A

Study question: Are there protein-level differences in circulating blood associated with acute nephritis in young patients, relative to controls without the condition?

Omics type: Mass spectrometry proteomics, data-independent acquisition

Study design: Cases were recruited prospectively at admission to hospital for nephritis and deliberately stratified into three roughly equal age groups to spread age evenly across the case group. Controls were assembled opportunistically from an existing sample biobank. There was no matched recruitment protocol running alongside case collection. Both groups are skewed toward one sex and younger ages.



Module 3: case study B

Study question: Can we identify genomic variants associated with body size across dog breeds using public data from unrelated studies?

Omics type: Whole genome sequencing

Study design: Whole genome sequencing data from 500 dogs across 134 breeds were aggregated from multiple public archives. No individual-level phenotypic measures were available other than reported breed identities. Phenotypic data were then constructed by reporting the median score for breeds using standards published by the American Kennel Club. Quantitative traits were labelled with a raw median value from the AKC, while binary traits were coded as 1 for trait present and 2 for trait absent.



Module 3: case study C

Study question: Can we identify differentially expressed genes across individual cell types in healthy individuals versus individuals presenting a pro-inflammatory condition?

Omics type: Single-cell RNA sequencing

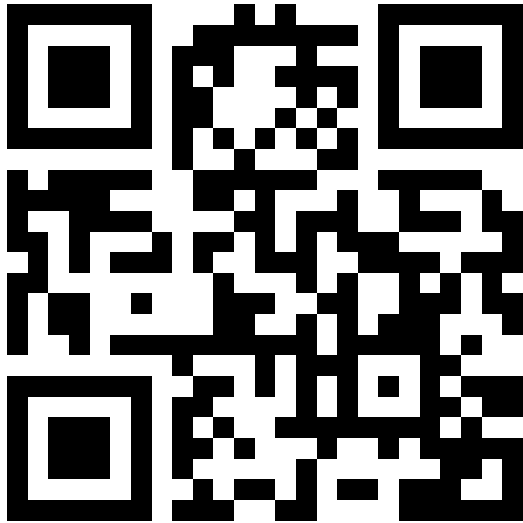
Study design: Single-cell RNA sequencing was collected from blood samples and enriched for circulating immune cells. Donors consist of two healthy individuals in the control group and two individuals presenting pro-inflammatory disease symptoms. The single-cell RNA sequencing data has undergone QC, preprocessing, and annotation for a total of 18 cell subtypes across T cells, natural killer cells, monocytes, and dendritic cells.

Wrap up

Recap

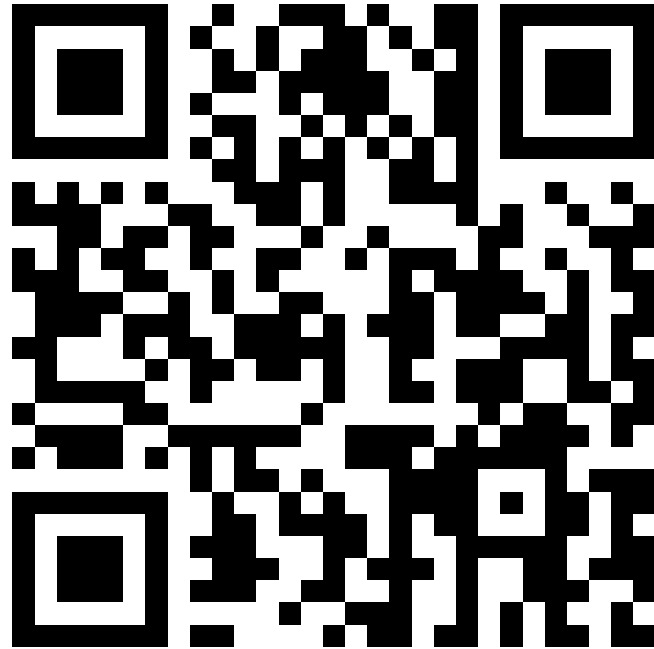
1. Match your chosen molecular layer to the biology driving your phenotype.
2. Treat design as a series of decisions with deadlines.
3. Design problems are cheap to fix early on and very expensive late.
4. Record every variable that could explain structure in your data.
5. Frame what your study is designed for before looking at your results.

Contact SIH



- Get help with your research: sih.tools/request
- Use our bioinformatics tools: sih.tools/bio
- Attend our training: sih.tools/training
- Use research compute: sih.tools/research-compute-guides

Survey



sih.tools/bio101-survey-2026

**This is our first time running this workshop, what
can we improve on?**